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Image-capturing apparatus, control method of image-capturing apparatus, display device, and image-capturing system

Abstract

An image-capturing apparatus includes an image-capturing device; a detecting unit configured to detect a facial direction of a user with respect to the image-capturing apparatus; a setting unit configured to set a clipping range in each frame image in a moving image that is image-captured by the image-capturing device, on the basis of the facial direction that is detected; and a generating unit that is configured to generate a clipped moving image from the clipping range. The generating unit generates the clipped moving image by changing the clipping range set with respect to a frame image in an action period, during which the user performs an action of directing a face toward a display device communicably connected to the image-capturing apparatus, to the clipping range set with respect to a frame image from before the action period starting.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an image-capturing apparatus, a control method of the image-capturing apparatus, a display device, and an image-capturing system.

Description of the Related Art

(2) Conventionally, image-capturing by a camera requires a camera operator to continuously direct the camera in an image-capturing direction. Accordingly, the camera operator may not be able to handle tasks other than the act of image-capturing, due to being preoccupied by image-capturing operations, and may not be able to concentrate on the experience of being in that place, due to being made to concentrate on image-capturing.

(3) For example, with respect to image-capturing operations, there is a problem in that a parent who is a camera operator will not be able to play with his/her children while performing image-capturing of the children, and will not be able to perform image-capturing while playing with the children.

(4) Also, with respect to concentration on image-capturing, there is a problem in that a camera operator who is a sports spectator will not be able to cheer while performing image-capturing, or

remember what happened in the game, and will not be able to perform image-capturing while concentrating on watching the game. In the same way, there is a problem in that a camera operator who is performing image-capturing while on a group tour will not be able to experience sensations on the same level as other members, and image-capturing will tend to be neglected if priority is given to sensing the experience.

(5) One method of solving such a problem is to fix an action camera on the head using a head-fixation accessory, and perform image-capturing in a direction being observed, thereby performing image-capturing without the camera operator being preoccupied by image-capturing operations. There also is a method in which image-capturing is performed over a wide range using an omnidirectional-shooting camera so as to concentrate on the experience while participating in the experience, and then clipping and editing necessary picture portions from the omnidirectional picture that has been image-captured after the experience, thereby preserving a picture of what has been experienced.

(6) However, the method of the former requires a troublesome act of wearing a head-fixation accessory to which a main unit of an action camera **901** has been fixed, such as illustrated in FIG. 27A, on the head. Also, when the camera operator wears the action camera **901** on the head using a head-fixation accessory **902** such as illustrated in FIG. 27B, there are problems such as the appearance being unflattering, the hair of the camera operator being mussed up, and so forth. Further, the camera operator may be distracted by the presence of the head-fixation accessory **902** and the action camera **901** worn on the head due to the weight and so forth thereof, and be concerned of how others are viewing his/her appearance. Accordingly, there have been problems that in a state such as illustrated in FIG. 27B, the camera operator may not be able to concentrate on the experience, and may feel resistance to the state illustrated in FIG. 27B itself to where image-capturing itself seems torturous.

(7) Conversely, the method of the latter requires a series of work such as image conversion, specifying clipping positions, and so forth. For example, there is known an omnidirectional-shooting camera **903** that includes a lens **904** and a shooting button **905**, as illustrated in FIG. 28. The lens **904** is one of a pair of fisheye lenses for omnidirectional shooting, configured on both faces of a housing of the omnidirectional-shooting camera **903**, and the omnidirectional-shooting camera **903** performs omnidirectional shooting using this pair of fisheye lenses. That is to say, omnidirectional shooting is performed by compositing projected images of this pair of fisheye lenses.

(8) FIGS. 29A to 29C are diagrams illustrating an example of conversion work performed on a picture that is image-captured by the omnidirectional-shooting camera **903**.

(9) FIG. 29A is an example of a picture obtained by omnidirectional shooting using the omnidirectional-shooting camera **903**, and includes a camera operator **906**, a child **907**, and a tree **908**, which are subjects. This picture is picture from a hemispherical optical system obtained by compositing projected images of the pair of fisheye lenses, and accordingly the camera operator **906** is greatly deformed. The torso portion of the child **907** which is the subject that the camera operator **906** was trying to perform image-capturing of is greatly deformed to the right and left, and spread out, since the torso portion was at the peripheral portion of the hemispherical optical system. Meanwhile, image-capturing of the tree **908** is performed with little distortion, due to being a subject situated to a front side of the lens **904**.

(10) In order to create a picture having a field of view like people usually see, out of the picture in FIG. 29A, part of the picture needs to be clipped and subjected to planar transform and displayed.

(11) FIG. 29B is a picture in which the picture positioned to the front of the lens **904** has been obtained by clipping, with respect the picture in FIG. 29A. The picture in FIG. 29B is a field of view like people usually see, with the tree **908** in the middle of the picture. However, the child **907** that is what the camera operator **906** was trying to perform image-capturing of is not included in FIG. 29B, and accordingly the clipping position needs to be changed. Here, the clipping position

specifically needs to be changed leftward and downward by 30° from the tree 908 situated to the front in FIG. 29A. A picture obtained by performing this clipping work and then performing planar transform and displaying is illustrated in FIG. 29C. In this way, the work of clipping necessary portions and performing planar transform needs to be performed in order to obtain the picture in FIG. 29C that the camera operator was trying to perform image-capturing for, from the picture in FIG. 29A. Accordingly, while the camera operator can concentrate on the experience while experiencing (while performing image-capturing), there is a problem that the amount of work thereafter is massive.

(12) Also, when a camera operator is shooting moving images with a camera and shifts his/her line of sight from the camera to a mobile terminal such as a smartphone or the like, in order to operate the smartphone, confirmation of the state of image-capturing at the camera is difficult. It is preferable for the camera operator to be able to continuously perform image-capturing of desired pictures even when temporarily directing his/her eyes toward a smartphone.

(13) Japanese Patent Application Publication No. 2020-150355 discloses technology in which, when a user is walking while operating a mobile terminal, and an abnormality is detected in the walking of the user, audio data and image data acquired for a predetermined amount of time after the abnormality occurs is transmitted to a managing server.

(14) In a case of a camera operator performing image-capturing of a moving image with a camera, and directing his/her line of sight away from a subject to a smartphone or the like, continuously performing image-capturing of desired pictures with the camera is difficult even when various types of information are received from the smartphone, unless processing for adjusting the image-capturing range and so forth is performed at the camera side.

SUMMARY OF THE INVENTION

(15) The present invention provides technology whereby image-capturing of desired pictures can be performed even when a user directs his/her line of sight away from the direction of a subject while performing image-capturing of a moving image.

(16) An image-capturing apparatus includes an image-capturing device; and at least one memory and at least one processor which function as: a detecting unit configured to detect a facial direction of a user with respect to the image-capturing apparatus; a setting unit configured to set a clipping range in each frame image in a moving image that is image-captured by the image-capturing device, on the basis of the facial direction that is detected; and a generating unit that is configured to generate a clipped moving image from the clipping range, wherein the generating unit generates the clipped moving image by changing the clipping range set with respect to a frame image in an action period, during which the user performs an action of directing a face toward a display device communicably connected to the image-capturing apparatus, to the clipping range set with respect to a frame image from before the action period starting.

(17) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a perspective view of a camera main unit including a shooting and detecting unit serving as an image-capturing apparatus according to a first embodiment;

(2) FIG. 1B is a diagram illustrating a user wearing the camera main unit;

(3) FIG. 1C is a diagram illustrating a battery unit of the camera main unit, as viewed from a rear side in FIG. 1A;

(4) FIG. 1D is an external view of a display device serving as mobile equipment according to the first embodiment, configured separately from the camera main unit;

- (5) FIG. 2A is a diagram viewing the shooting and detecting unit from a front side;
- (6) FIG. 2B is a diagram illustrating forms of band portions of connecting portions of the camera main unit;
- (7) FIG. 2C is a diagram illustrating the shooting and detecting unit as viewed from the rear side;
- (8) FIG. 2D is a diagram illustrating the shooting and detecting unit as viewed from above;
- (9) FIG. 2E is a diagram illustrating a configuration of an infrared detection processing device that is disposed in a lower portion of a facial direction detection window of the camera main unit, inside the shooting and detecting unit;
- (10) FIG. 2F is a diagram illustrating a state in which the user is wearing the camera main unit, as viewed from a left side of the user;
- (11) FIGS. 3A to 3C are diagrams for describing details of the battery unit;
- (12) FIG. 4 is a functional block diagram of the camera main unit according to the first embodiment;
- (13) FIG. 5 is a block diagram illustrating a hardware configuration of the camera main unit;
- (14) FIG. 6 is a block diagram illustrating a hardware configuration of the display device;
- (15) FIG. 7A is a flowchart showing an overview of imaging recording processing according to the first embodiment, which is executed at the camera main unit and the display device;
- (16) FIG. 7B is a flowchart of a subroutine for preparatory operation processing of step **S100** in FIG. 7A according to the first embodiment;
- (17) FIG. 7C is a flowchart of a subroutine for facial direction detection processing of step **S200** in FIG. 7A according to the first embodiment;
- (18) FIG. 7D is a flowchart of a subroutine for recording direction and range deciding processing of step **S300** in FIG. 7A according to the first embodiment;
- (19) FIG. 7E is a flowchart of a subroutine for recording range developing processing of step **S500** in FIG. 7A according to the first embodiment;
- (20) FIG. 7F is a diagram for describing processing of steps **S200** to **S500** in FIG. 7A in a moving image mode;
- (21) FIG. 8A is a diagram illustrating a picture of the user as seen from the facial direction detection window;
- (22) FIG. 8B is a diagram illustrating a case in which fluorescent lamps of a room show up in a picture of the user as a background, as seen from the facial direction detection window;
- (23) FIG. 8C is a diagram illustrating a picture in a case in which the user and the fluorescent lamps in the background, illustrated in FIG. 8B, are imaged by a sensor of an infrared detection processing device via the facial direction detection window, in a state in which infrared light-emitting diodes (LED) of the infrared detection processing device are not lit;
- (24) FIG. 8D is a diagram illustrating a picture in a case in which the user and the background fluorescent lamps illustrated in FIG. 8B are imaged by the sensor of the infrared detection processing device via the facial direction detection window, in a state in which the infrared LEDs are lit;
- (25) FIG. 8E is a diagram illustrating a differential picture computed from the pictures in FIGS. 8C and 8D;
- (26) FIG. 8F is a diagram illustrating a case in which shades of the differential picture in FIG. 8E have been adjusted in scale, in accordance with light intensity of reflected light from the infrared light cast on face and neck portions of the user;
- (27) FIG. 8G is a diagram in which signs denoting portions of the body of the user, and double circle and black dot symbols, respectively indicating a neck base portion position and a chin tip portion position, are overlaid on FIG. 8F;
- (28) FIG. 8H is a diagram illustrating a differential picture computed by the same method as in FIG. 8E, when the face of the user is directed in a right direction;
- (29) FIG. 8I is a diagram in which the double circle and black dot symbols, respectively indicating

the neck base portion position and the position of the chin tip portion, are overlaid on FIG. 8H;

(30) FIG. 8J is a diagram illustrating a picture of the user as seen from the facial direction detection window, in a case in which the face of the user is directed 33° upward from a horizontal direction;

(31) FIG. 8K is a diagram illustrating a picture of the user in which the double circle and black dot symbols, respectively indicating the neck base portion position and the position of the chin tip portion, are overlaid on a differential picture computed by the same method as in FIG. 8E, in a case in which the face of the user is directed 33° upward from the horizontal direction;

(32) FIGS. 9A to 9E are timing charts showing lighting timings of the infrared LEDs;

(33) FIGS. 10A to 10D are diagrams for describing movement of the face of the user in an up-down direction;

(34) FIG. 11A is a diagram illustrating an aiming field of view in an ultrawide angle picture that is image-captured by a shooting unit of the camera main unit, in a case in which the user is facing toward the front;

(35) FIG. 11B is a diagram illustrating a picture of the aiming field of view in FIG. 11A, obtained by clipping from the ultrawide angle picture;

(36) FIG. 11C is a diagram illustrating the aiming field of view in the ultrawide angle picture in a case in which the user is observing a subject A;

(37) FIG. 11D is a diagram illustrating a picture in which a picture of the aiming field of view in FIG. 11C, obtained by clipping from the ultrawide angle picture, has been subjected to correction of distortion and blurring;

(38) FIG. 11E is a diagram illustrating the aiming field of view in the ultrawide angle picture in a case in which the user is observing the subject A, at an angle of view settings value that is smaller than that in FIG. 11C;

(39) FIG. 11F is a diagram illustrating a picture in which a picture of the aiming field of view in FIG. 11E, obtained by clipping from the ultrawide angle picture, has been subjected to correction of distortion and blurring;

(40) FIG. 12A is a diagram illustrating an example of an aiming field of view in an ultrawide angle picture;

(41) FIG. 12B is a diagram illustrating an example of an aiming field of view in the ultrawide angle picture, in which the angle of view settings value is the same as that in FIG. 12A, but an observation direction is different;

(42) FIG. 12C is a diagram illustrating another example of an aiming field of view in the ultrawide angle picture, in which the angle of view settings value is the same as that in FIG. 12A, but the observation direction is different;

(43) FIG. 12D is a diagram illustrating an example of an aiming field of view in the ultrawide angle picture, in which the observation direction is the same as that in FIG. 12C, but the angle of view settings value is smaller;

(44) FIG. 12E is a diagram illustrating an example of imparting a reserve region to a perimeter of the aiming field of view illustrated in FIG. 12A;

(45) FIG. 12F is a diagram illustrating an example of imparting a reserve region of an image stabilization level, which is the same as that of the reserve region in FIG. 12E, to the perimeter of the aiming field of view illustrated in FIG. 12B;

(46) FIG. 12G is a diagram illustrating an example of imparting a reserve region of an image stabilization level, which is the same as that of the reserve region in FIG. 12E, to the perimeter of the aiming field of view illustrated in FIG. 12D;

(47) FIG. 13 is a diagram illustrating a menu screen for various types of settings for the moving image mode, which is displayed on a display unit of the display device before performing image-capturing by the camera main unit;

(48) FIG. 14 is a flowchart of a subroutine for primary recording processing of step S600 in FIG. 7A;

- (49) FIG. 15 is a diagram illustrating a data structure of a picture file generated by the primary recording processing;
- (50) FIG. 16 is a flowchart of a subroutine for transfer processing to the display device of step S700 in FIG. 7A;
- (51) FIG. 17 is a flowchart of a subroutine for optical correction processing of step S800 in FIG. 7A;
- (52) FIGS. 18A to 18F are diagrams for describing a case of implementing distortion aberration correction in step S803 of FIG. 17;
- (53) FIG. 19 is a flowchart of a subroutine for image stabilization processing of step S900 in FIG. 7A;
- (54) FIG. 20 is a flowchart of preparatory operation processing according to a second embodiment;
- (55) FIG. 21 is a flowchart of processing according to the second embodiment, performed by the display device 800;
- (56) FIG. 22 is a flowchart for determining an action period for viewing the display device;
- (57) FIG. 23 is a diagram exemplifying frame management information;
- (58) FIG. 24 is a diagram for describing states of flags in the action period;
- (59) FIG. 25 is a flowchart of recording range developing processing according to the second embodiment;
- (60) FIG. 26 is a flowchart of recording range finalization developing processing;
- (61) FIGS. 27A and 27B are diagrams illustrating a configuration example of a camera fixed to a head using a head-fixation accessory according to conventional art;
- (62) FIG. 28 is a diagram illustrating a configuration example of an omnidirectional shooting camera according to conventional art; and
- (63) FIGS. 29A to 29C are diagrams illustrating an example of conversion work performed on a picture that is image-captured by the omnidirectional-shooting camera in FIG. 28.

DESCRIPTION OF THE EMBODIMENTS

(64) Preferred embodiments of the present invention will be described below in detail with reference to the attached drawings.

First Embodiment

- (65) FIGS. 1A to 1D are diagrams for describing a camera system made up of a camera main unit 1 including a shooting and detecting unit 10 serving as an image-capturing apparatus according to the present embodiment, and a display device 800 that is configured separately therefrom. Note that while an example is illustrated in the present embodiment regarding a case in which the camera main unit 1 and the display device 800 are separate, but these may be configured integrally. Also, a user who is wearing the camera main unit 1 on his/her neck will be referred to as “user”.
- (66) FIG. 1A is an external view of the camera main unit 1.
- (67) In FIG. 1A, the camera main unit 1 includes the shooting and detecting unit 10, a battery unit 90, and connecting portions 80 that connect the shooting and detecting unit 10 and the battery unit 90 (power supply means).
- (68) The shooting and detecting unit 10 includes a facial direction detection window 13, a start switch 14, a stop switch 15, an image-capturing lens 16, a light-emitting diode (LED) 17, and microphones 19L and 19R.
- (69) The facial direction detection window 13 transmits infrared light cast from an infrared LED lighting circuit 21 (infrared light illuminating means, see FIG. 5), built into the shooting and detecting unit 10, for detecting positions of parts of the face of the user, and light reflected therefrom.
- (70) The start switch 14 is a switch for starting image-capturing.
- (71) The stop switch 15 is a switch for stopping image-capturing.
- (72) The image-capturing lens 16 guides light to be image-captured to a solid-state image-capturing device 42 (FIG. 5) inside the shooting and detecting unit 10.

(73) The LED **17** is an LED that indicates that image-capturing is being performed, gives warnings, and so forth.

(74) The microphones **19L** and **19R** are microphones for picking up sound from the vicinity, with the microphone **19L** picking up sound from the vicinity to the left side of the user (the right side in FIG. **1A**) and the microphone **19R** picking up sound from the vicinity to the right side of the user (the left side in FIG. **1A**).

(75) FIG. **1B** is a diagram illustrating the user wearing the camera main unit **1**.

(76) When wearing the camera main unit **1** such that the battery unit **90** is situated at the back side of the user, and the shooting and detecting unit **10** is situated at the front side of the body of the user, the camera main unit **1** is biased toward the chest of the user by the connecting portions **80** connected to both ends of the shooting and detecting unit **10** near the right and left end portions thereof, and thus supported. Accordingly, the shooting and detecting unit **10** is positioned around the front of the collarbones of the user. At this time, the facial direction detection window **13** is positioned below the chin of the user. An infrared light condensing lens **26**, which will be described later with reference to FIG. **2E**, is disposed within the facial direction detection window **13**. An optical axis of the image-capturing lens **16** (image-capturing optical axis) and an optical axis of the infrared light condensing lens **26** (detecting optical axis) are directed in different directions, with a facial direction detecting unit **20** (a detecting unit), which will be described later, detecting an observation direction of the user, from the positions of various parts of the face. Thus, image capturing in this observation direction by a shooting unit **40** (an image-capturing device), which will be described later, is enabled.

(77) Methods for adjusting a setting position, and so forth, in accordance with individual difference in body figures, difference in clothing, and so forth, will be described later.

(78) Also, disposing the shooting and detecting unit **10** on the front side of the body and the battery unit **90** on the back side in this way disperses weight, and has effects of reducing user fatigue and suppressing shifting under centrifugal force and so forth when the user moves.

(79) Note that while an example is given in the present embodiment in which the shooting and detecting unit **10** is worn so as to be positioned around the front of the collarbones of the user, this is not restrictive. That is to say, the camera main unit **1** may be worn at any part of the body of the user, other than the head, as long as the camera main unit **1** can detect the observation direction of the user by the facial direction detecting unit **20**, and can also perform image-capturing in this observation direction by the shooting unit **40**.

(80) FIG. **1C** is a diagram illustrating the battery unit **90**, as viewed from a rear side in FIG. **1A**.

(81) In FIG. **1C**, the battery unit **90** includes a charging cable insertion port **91**, adjustment buttons **92L** and **92R**, and a backbone accommodation notch **93**.

(82) The charging cable insertion port **91** is an insertion port for a charging cable that is omitted from illustration. Batteries **94** that are inside are charged, and electric power is supplied to the shooting and detecting unit **10**, from an external electric power supply via this charging cable.

(83) The adjustment buttons **92L** and **92R** are adjustment buttons for adjusting the length of band portions **82L** and **82R** of the connecting portions **80**. The adjustment button **92L** is a button for adjusting the band portion **82L** to the left side, and the adjustment button **92R** is a button for adjusting the band portion **82R** to the right side. Note that while the lengths of the band portions **82L** and **82R** are each independently adjusted by the respective adjustment buttons **92L** and **92R** in the present embodiment, an arrangement may be made in which the lengths of the band portions **82L** and **82R** are adjusted at the same time by a single button. Hereinafter, the band portions **82L** and **82R** will be collectively referred to as “band portions **82**”.

(84) The backbone accommodation notch **93** is a notch corresponding to the profile of the backbone, so that the battery unit **90** does not press against the backbone portion of the user. Avoiding the protruding portion of the human backbone reduces discomfort in wearing, and at the same time prevents the main unit from moving laterally while being worn.

(85) FIG. 1D is an external view of the display device **800** serving as mobile equipment in the first embodiment, configured separately from the camera main unit **1**.

(86) In FIG. 1D, the display device **800** includes a button A **802**, a display unit **803**, a button B **804**, a front camera **805**, a face sensor **806**, an angular velocity sensor **807**, and an acceleration sensor **808**. Also provided is a wireless local area network (LAN) that enables high-speed connection to the camera main unit **1**, although this is omitted from illustration in FIG. 1D.

(87) The button A **802** is a button that has functions of a power button of the display device **800**, accepts on/off operations by long-press operations, and accepts instructions regarding other processing timings by short-press operations.

(88) The display unit **803** enables pictures image-captured by the camera main unit **1** to be confirmed, and menu screens necessary for settings to be displayed. In the present embodiment, a transparent touch sensor may be provided on a top face of the display unit **803**, to accept touch operations on a screen being displayed (e.g., a menu screen).

(89) The button B **804** is a button that functions as a calibration button used in calibration processing.

(90) The front camera **805** is a camera that is capable of performing image-capturing of a person who is observing the display device **800**.

(91) The face sensor **806** detects the facial forms of the person who is observing the display device **800**, and the observation direction thereof. The specific structure of the face sensor **806** is not limited in particular, and can be implemented by various types of sensors, such as a structured light sensor, a time-of-flight (ToF) sensor, a millimeter-wave radar device, or the like, for example.

(92) The angular velocity sensor **807** is indicated by dotted lines, used to represent objects that are hidden from sight within the display device **800**. A triaxial gyro sensor, for the three axial directions of X, Y, and Z, is installed in the display device **800** according to the present embodiment, in order to also function as a calibrator, which will be described later.

(93) The acceleration sensor **808** detects the attitude of the display device **800**.

(94) Note that a common smartphone is used for the display device **800** according to the present embodiment, and the camera system according to the present invention can be implemented by arranging firmware in the smartphone to correspond to firmware at the camera main unit **1** side. Note however, that the camera system according to the present invention can also be implemented by arranging the firmware at the camera main unit **1** side to correspond to an application or operating system (OS) of the smartphone serving as the display device **800**.

(95) FIGS. 2A to 2F are diagrams for describing the shooting and detecting unit **10** in detail. In the Figures hereinafter, portions that have already been described will be denoted by the same numerals, thereby indicating that the functions thereof are the same, and description in the present specification will be omitted.

(96) FIG. 2A is a diagram viewing the shooting and detecting unit **10** from a front side.

(97) The connecting portions **80** connect to the shooting and detecting unit **10** at a right-side connecting portion **80R** situated on the right side of the body of the user (to the left side in FIG. 2A) and a left-side connecting portion **80L** situated on the left side of the body of the user (to the right side in FIG. 2A). In detail, the connecting portions **80** are each made up of an angle holding portion **81** that is made of a hard material, for holding an angle as to the shooting and detecting unit **10**, and a band portion **82**. That is to say, the right-side connecting portion **80R** includes an angle holding portion **81R** and a band portion **82R**, and the left-side connecting portion **80L** includes an angle holding portion **81L** and a band portion **82L**.

(98) FIG. 2B is a diagram illustrating forms of the band portions **82** of the connecting portions **80**. The angle holding portions **81** are illustrated in outline only in FIG. 2B, in order to show the forms of the band portions **82**.

(99) The band portions **82** include connecting faces **83** and an electric cable **84**.

(100) The connecting faces **83** are connecting faces of the angle holding portions **81** and the band

portions **82**, and have cross-sectional forms that are not true circles, elliptical forms herein. Hereinafter, the connecting faces **83** disposed in lateral symmetry to each other on the right side of the body of the user (to the left side in FIG. 2B) and on the left side of the body of the user (to the right side in FIG. 2B) when wearing the camera main unit **1** will be referred to as “right-side connecting face **83R**” and “left-side connecting face **83L**”. The right-side connecting face **83R** and the left-side connecting face **83L** are arranged in a downward-splayed form. That is to say, the distance between the right-side connecting face **83R** and the left-side connecting face **83L** becomes smaller the further in the upward direction from a lower side in FIG. 2B. Thus, when the user wears the camera main unit **1**, major axes of the connecting faces **83** of the connecting portions **80** are arranged in directions following the body of the user, which is advantageous in that the band portions **82** are comfortable when coming into contact with the body of the user, and also the shooting and detecting unit **10** does not move in lateral and forward/rearward directions.

(101) The electric cable **84** (electric power supply means) is a cable that is routed through inside of the band portion **82L**, and electrically connects the battery unit **90** and the shooting and detecting unit **10**. The electric cable **84** connects the electric power supply of the battery unit **90** to the shooting and detecting unit **10**, and externally exchanges electric signals.

(102) FIG. 2C is a diagram illustrating the shooting and detecting unit **10** as viewed from a rear side. FIG. 2C is a diagram viewed from the side in contact with the body of the user, i.e., from the opposite side as to FIG. 2A, and accordingly the positional relation between the right-side connecting portion **80R** and the left-side connecting portion **80L** is reverse from that in FIG. 2A.

(103) The shooting and detecting unit **10** includes a power switch **11**, an image-capturing mode switch **12**, and chest connecting pads **18**, on the rear side thereof.

(104) The power switch **11** is a power switch that switches the power of the camera main unit **1** on/off. The power switch **11** according to the present embodiment is a sliding lever switch, but is not limited to this. For example, the power switch **11** may be a pushbutton switch, or may be a switch that is integrally configured with a slide cover (omitted from illustration) of the image-capturing lens **16**.

(105) The image-capturing mode switch **12** (changing means) is a switch for changing the image-capturing mode, and can change modes relating to image-capturing. In the present embodiment, besides switching to still-image mode and moving image mode, the image-capturing mode switch **12** is also capable of switching to a later-described Pri setting mode, in which settings are performed using the display device **800**. In the present embodiment, the image-capturing mode switch **12** is a sliding lever switch, by which one of “Photo”, “Normal”, and “Pri”, illustrated in FIG. 2C, can be selected by sliding a lever. The image-capturing mode transitions to still-image mode by sliding to “Photo”, transitions to moving image mode by sliding to “Normal”, and transitions to Pri setting mode by sliding to “Pri”. Note that the image-capturing mode switch **12** is not limited to the form in the present embodiment, as long as it is a switch that is capable of changing image-capturing modes. For example, the image-capturing mode switch **12** may be made up of three buttons of “Photo”, “Normal”, and “Pri”.

(106) The chest connecting pads **18** (fixing means) are parts that come into contact with the body of the user when the shooting and detecting unit **10** is biased against the body of the user. As illustrated in FIG. 2A, the shooting and detecting unit **10** is formed such that when worn, the full length in the lateral (right-and-left) direction is longer than the full length in the upright (up-and-down) direction, and the chest connecting pads **18** are disposed near the right and left end portions of the shooting and detecting unit **10**. This arrangement enables lateral rotational blurring to be suppressed from occurring while performing image-capturing with the camera main unit **1**. Also, placing the chest connecting pads **18** at the chest enables the power switch **11** and the image-capturing mode switch **12** to be kept from coming into contact with the body. The chest connecting pads **18** further serves to prevent heat of the shooting and detecting unit **10** that has become hot from extended periods of image-capturing, from being conveyed to the body of the user, and to

adjust the angle of the shooting and detecting unit **10**.

(107) FIG. 2D is a diagram illustrating the shooting and detecting unit **10** as viewed from above.

(108) As illustrated in FIG. 2D, the facial direction detection window **13** is provided at the middle portion of a top face of the shooting and detecting unit **10**, and the chest connecting pads **18** protrude from the shooting and detecting unit **10**.

(109) FIG. 2E is a diagram illustrating a configuration of an infrared detection processing device **27** that is disposed within the shooting and detecting unit **10**, below the facial direction detection window **13**.

(110) The infrared detection processing device **27** includes infrared LEDs **22** and the infrared light condensing lens **26**.

(111) The infrared LEDs **22** cast infrared light **23** (FIG. 5) toward the user.

(112) The infrared light condensing lens **26** is a lens that images reflected light **25** (FIG. 5), which is reflected at the user and returns when casting the infrared light **23** from the infrared LEDs **22**, to a sensor (omitted from illustration) of the infrared detection processing device **27**.

(113) FIG. 2F is a diagram illustrating a state in which the user is wearing the camera main unit **1**, as viewed from a left side.

(114) An angle adjustment button **85L** is a button provided to the angle holding portion **81L**, and is used for adjusting the angle of the shooting and detecting unit **10**. Note that although omitted from illustration in FIG. 2F, an angle adjustment button **85R** is provided within the angle holding portion **81R** on the opposite side face as well, at a position that is symmetrical with the angle adjustment button **85L**. Hereinafter, when collectively referring to the angle adjustment buttons **85R** and **85L**, these will be referred to as “angle adjustment buttons **85**”.

(115) The angle adjustment buttons **85** are at positions that would be visible in FIGS. 2A, 2C, and 2D as well, but have been omitted from illustration to simplify description.

(116) The user can change the angle between the shooting and detecting unit **10** and the angle holding portions **81**, by moving the angle holding portions **81** in the up and down directions in FIG. 2F while pressing the angle adjustment buttons **85**. Also, the protrusion angle of the chest connecting pads **18** is adjustable. The orientation of the image-capturing lens **16** of the shooting and detecting unit **10** can be adjusted to be horizontal with respect to individual differences in the form of the chest position of users, by the action of these two angle changing members (angle adjustment buttons **85** and chest connecting pads **18**).

(117) FIGS. 3A to 3C are diagrams for describing details of the battery unit **90**.

(118) FIG. 3A is a diagram of the battery unit **90** from the rear face thereof, with parts illustrated in outline.

(119) As illustrated in FIG. 3A, in order to balance the weight of the battery unit **90**, the two of a left battery **94L** and a right battery **94R** (hereinafter also collectively referred to as “batteries **94**”) are symmetrically installed within. Symmetrically disposing the batteries **94** as to the middle portion of the battery unit **90** in this way balances the weight laterally, and prevents positional deviation of the camera main unit **1**. Note that a configuration may be made for the battery unit **90** in which only one battery is installed.

(120) FIG. 3B is a diagram of the battery unit **90** as viewed from above. The batteries **94** are illustrated in outline in FIG. 3B as well.

(121) The relation between the backbone accommodation notch **93** and the batteries **94** can be understood from FIG. 3B. Disposing the batteries **94** symmetrically on both sides of the backbone accommodation notch **93** enables the user to wear the battery unit **90**, which is relatively heavy, with little load.

(122) FIG. 3C is a diagram viewing the battery unit **90** from the rear side thereof. FIG. 3C is a diagram viewed from a side that comes into contact with the body of the user, i.e., from the opposite side as to FIG. 3A.

(123) As illustrated in FIG. 3C, the backbone accommodation notch **93** is provided at the middle,

following the backbone of the user.

(124) FIG. 4 is a functional block diagram of the camera main unit 1. While details will be described later, a general flow of processing that is executed at the camera main unit 1 will be described here with reference to FIG. 4.

(125) In FIG. 4, the camera main unit 1 includes the facial direction detecting unit 20, a recording direction and angle-of-view deciding unit 30, the shooting unit 40, an image clipping and developing processing unit 50, a primary recording unit 60, a transmission unit 70, and an other-controlling unit 111. These functional blocks are executed under control of a general control central processing unit (CPU) 101 (FIG. 5) that performs general control of the camera main unit 1.

(126) The facial direction detecting unit 20 (observation direction detecting means) is a functional block executed by the infrared LEDs 22, the infrared detection processing device 27, and so forth, which have been described earlier, detects a facial direction and infers the observation direction, which is then handed to the recording direction and angle-of-view deciding unit 30.

(127) The recording direction and angle-of-view deciding unit 30 (recording direction deciding means) performs various types of computation on the basis of the observation direction inferred by the facial direction detecting unit 20, decides information of a position and a range for clipping a picture from the shooting unit 40, and hands this information to the image clipping and developing processing unit 50.

(128) The shooting unit 40 converts the light from the subject into a picture, and hands this picture to the image clipping and developing processing unit 50.

(129) The image clipping and developing processing unit 50 (developing means) clips the picture from the shooting unit 40 using the information from the recording direction and angle-of-view deciding unit 30, and performs developing thereof, and thereby hands just a picture, taken in the direction that the user is looking, to the primary recording unit 60.

(130) The primary recording unit 60 is a functional block made up of primary memory 103 (FIG. 5) and so forth. The primary recording unit 60 records picture information, and hands the picture information to the transmission unit 70 at a timing when necessary.

(131) The transmission unit 70 (picture output means) wirelessly connects to the display device 800 (FIG. 1D), a calibrator 850, and a simplified display device 900, which are communication partners decided in advance, and performs communication therewith.

(132) The display device 800 is a display device that is capable of connection to the transmission unit 70 by a wireless LAN that is capable of high-speed connection (hereinafter, referred to as "high-speed wireless"). Although wireless communication conforming to the IEEE 802.11ax (Wi-Fi 6) standard is used for the high-speed wireless in the present embodiment, but wireless communication conforming to other standards, such as the Wi-Fi 4 standard or the Wi-Fi 5 standard, for example, may be used. Also, the display device 800 may be equipment developed specifically for the camera main unit 1, or may be a common smartphone, tablet terminal, or the like.

(133) Note that connection between the transmission unit 70 and the display device 800 may use small power wireless, or connection may be made by both high-speed wireless and small power wireless, or connecting by switching therebetween. In the present embodiment, large amounts of data, such as picture files of moving image pictures or the like, which will be described later, are transmitted by high-speed wireless, and lightweight data or data regarding which taking time is not a problem, are transmitted by small power wireless. Although Bluetooth is used here for small power wireless in the present embodiment, other small power wireless may be used, such as near-field communication, short-range communication, or the like.

(134) The calibrator 850 is equipment for performing initial settings or individual settings of the camera main unit 1, and is equipment that is capable of connection to the transmission unit 70 by high-speed wireless, in the same way as the display device 800. The calibrator 850 will be described in detail later. Also, the display device 800 may also have the functions of this calibrator

850.

(135) The simplified display device **900** is a display device that can only connect to the transmission unit **70** by small power wireless, for example.

(136) The simplified display device **900** is a display device that cannot transmit moving image pictures with the transmission unit **70** due to time-related constraints, but can transmit timing for starting/stopping image-capturing, image confirmation such as confirming composure, and so forth. Also, the simplified display device **900** may be equipment developed specifically for the camera main unit **1**, or may be a smartwatch or the like, in the same way as the display device **800**.

(137) FIG. **5** is a block diagram illustrating a hardware configuration of the camera main unit **1**. Configurations and functions described with reference to FIGS. **1A** to **1C** and so forth are denoted by the same numerals, and detailed description thereof will be omitted.

(138) In FIG. **5**, the camera main unit **1** includes the general control CPU **101**, the power switch **11**, the image-capturing mode switch **12**, the facial direction detection window **13**, the start switch **14**, the stop switch **15**, the image-capturing lens **16**, and the LED **17**.

(139) The camera main unit **1** also includes the infrared LED lighting circuit **21**, the infrared LEDs **22**, the infrared light condensing lens **26**, and the infrared detection processing device **27** that make up the facial direction detecting unit **20** (FIG. **4**).

(140) The camera main unit **1** also includes the shooting unit **40** (FIG. **4**) that is made up of an image-capturing driver **41**, the solid-state image-capturing device **42**, and an image-capturing signal processing circuit **43**, and the transmission unit **70** (FIG. **4**) that is made up of a small power wireless unit **71** and a high-speed wireless unit **72**.

(141) Note that while the camera main unit **1** only includes one shooting unit **40** in the present embodiment, arrangements may be made in which two or more shooting units **40** are provided to perform image-capturing of three-dimensional pictures, to perform image-capturing of pictures over a wider angle of view than that which can be acquired by a single shooting unit **40**, or to perform image-capturing in a plurality of directions.

(142) The camera main unit **1** also includes various types of memory, such as large-capacity nonvolatile memory **51**, built-in nonvolatile memory **102**, the primary memory **103**, and so forth.

(143) Further, the camera main unit **1** includes an audio processing unit **104**, a speaker **105**, a vibrating member **106**, an angular velocity sensor **107**, an acceleration sensor **108**, and various types of switches **110**.

(144) The general control CPU **101** is connected to the power switch **11** described above with reference to FIG. **2C**, and so forth, and controls the camera main unit **1**. The recording direction and angle-of-view deciding unit **30**, the image clipping and developing processing unit **50**, and the other-controlling unit **111**, illustrated in FIG. **4**, are made up of the general control CPU **101** itself.

(145) The infrared LED lighting circuit **21** performs control of lighting and turning off the infrared LEDs **22** described above with reference to FIG. **2E**, and controls the infrared light **23** that is cast from the infrared LEDs **22** toward the user.

(146) The facial direction detection window **13** is made up of a visible light cutout filter, and can transmit very little visible light, but can sufficiently transmit the infrared light **23** and the reflected light **25** thereof that is light of the infrared region.

(147) The infrared light condensing lens **26** is a lens for collecting the reflected light **25**.

(148) The infrared detection processing device **27** (infrared light detecting means) has a sensor that detects the reflected light **25** collected by the infrared light condensing lens **26**. This sensor images the reflected light **25**, which has been collected, as a picture which is then converted into sensor data and handed to the general control CPU **101**.

(149) In a case in which the user is wearing the camera main unit **1** as illustrated in FIG. **1B**, the facial direction detection window **13** is situated below the chin of the user. Accordingly, the infrared light **23** cast from the infrared LED lighting circuit **21** is transmitted through the facial direction detection window **13** and is cast onto an infrared light illumination face **24**, which is

around the chin of the user, as illustrated in FIG. 5. Also, the infrared light 23 reflected at the infrared light illumination face 24 becomes the reflected light 25, is transmitted through the facial direction detection window 13, and is collected at the sensor in the infrared detection processing device 27 by the infrared light condensing lens 26.

(150) The various types of switches 110 are omitted from illustration in FIGS. 1A to 1C and so forth, and while details will be omitted from description, these are switches for executing functions that are unrelated to the present embodiment.

(151) The image-capturing driver 41 includes a timing generator and so forth, and generates and outputs various types of timing signals to various parts related to image-capturing, so as to perform image-capture driving.

(152) The solid-state image-capturing device 42 outputs signals obtained by photoelectric conversion of a subject image, projected from the image-capturing lens 16 described with reference to FIG. 1A, to the image-capturing signal processing circuit 43.

(153) The image-capturing signal processing circuit 43 outputs image-capturing data, generated by performing processing such as clamping and processing such as analog-to-digital (A/D) conversion on signals from the solid-state image-capturing device 42, to the general control CPU 101.

(154) The built-in nonvolatile memory 102 uses flash memory or the like, and stores activation programs for the general control CPU 101, and setting values for various types of program modes. In the present embodiment, the observation field of view (angle of view) can be changed, and the level of effect of image stabilization control can be set, and setting values thereof are recorded in the built-in nonvolatile memory 102 as well.

(155) The primary memory 103 is made up of random-access memory (RAM) or the like, and temporarily stores picture data being processed, temporarily stores computation results from the general control CPU 101, and so forth.

(156) The large-capacity nonvolatile memory 51 records and reads primary image data. Although description will be made in the present embodiment regarding a case in which the large-capacity nonvolatile memory 51 is semiconductor memory without a mounting/unmounting mechanism, for the sake of simplification of description, this is not restrictive. For example, the large-capacity nonvolatile memory 51 may be made up of a detachable recording medium such as a Secure Digital (SD) card or the like, and may be used in conjunction with the built-in nonvolatile memory 102.

(157) The small power wireless unit 71 performs exchange of data among the display device 800, the calibrator 850, and the simplified display device 900, by small power wireless.

(158) The high-speed wireless unit 72 performs exchange of data among the display device 800, the calibrator 850, and the simplified display device 900, by high-speed wireless.

(159) The audio processing unit 104 includes the microphone 19L on the right side in FIG. 1A and the microphone 19R on the left side in FIG. 1A, for collecting external sound (analog signals), and performs processing of the collected analog signals to generate audio signals.

(160) The LED 17, the speaker 105, and the vibrating member 106 respectively emit light, emit sound, and vibrate, thereby notifying or warning the user of the state of the camera main unit 1.

(161) The angular velocity sensor 107 is a sensor that uses a gyroscope and so forth, and detects movement of the camera main unit 1 itself as gyro data.

(162) The acceleration sensor 108 detects attitude of the shooting and detecting unit 10.

(163) Note that the angular velocity sensor 107 and the acceleration sensor 108 are built into the shooting and detecting unit 10, and that the angular velocity sensor 807 and the acceleration sensor 808, which are separate entities from these, are also provided in the display device 800 which will be described later.

(164) FIG. 6 is a block diagram illustrating a hardware configuration of the display device 800. Parts described with reference to FIG. 1D are denoted by the same signs for the sake of simplification of description, and description thereof will be omitted.

(165) In FIG. 6, the display device 800 includes a display device control unit 801, the button A 802,

the display unit **803**, the button **B 804**, the front camera **805**, the face sensor **806**, the angular velocity sensor **807**, the acceleration sensor **808**, an image-capturing signal processing circuit **809**, and various types of switches **811**.

(166) The display device **800** also includes built-in nonvolatile memory **812**, primary memory **813**, large-capacity nonvolatile memory **814**, a speaker **815**, a vibrating member **816**, an LED **817**, an audio processing unit **820**, a small power wireless unit **871**, and a high-speed wireless unit **872**.

(167) The display device control unit **801** is made up of a CPU, and is connected to the button **A 802**, the face sensor **806**, and so forth, described with reference to FIG. **1D**, and controls the display device **800**.

(168) The image-capturing signal processing circuit **809** bears functions equivalent to those of the image-capturing driver **41**, the solid-state image-capturing device **42**, and the image-capturing signal processing circuit **43** within the camera main unit **1**, but is not very important in the description of the present embodiment, and accordingly these are collectively described for the sake of simplification of description. Data output from the image-capturing signal processing circuit **809** is processed within the display device control unit **801**. The processing contents of this data will be described later.

(169) The various types of switches **811** are omitted from illustration in FIG. **1D**, and while details will be omitted from description, these are switches for executing functions that are unrelated to the present embodiment.

(170) The angular velocity sensor **807** is a sensor using a gyroscope and so forth, and detects movement of the display device **800** itself.

(171) The acceleration sensor **808** detects the attitude of the display device **800** itself.

(172) Note that the angular velocity sensor **807** and the acceleration sensor **808** are built into the display device **800**, as described above, and are separate entities from the angular velocity sensor **107** and the acceleration sensor **108** of the camera main unit **1** described above, although the functions are the same.

(173) The built-in nonvolatile memory **812** uses flash memory or the like, and stores activation programs for the display device control unit **801**, and setting values for various types of program modes.

(174) The primary memory **813** is made up of RAM or the like, and temporarily stores picture data being processed, temporarily stores computation results from the image-capturing signal processing circuit **809**, and so forth. In the present embodiment, while recording moving image pictures, gyro data detected by the angular velocity sensor **807** at the image-capturing time of each frame is stored in the primary memory **813** in an associated manner with each frame.

(175) The large-capacity nonvolatile memory **814** records and reads image data for the display device **800**. In the present embodiment, the large-capacity nonvolatile memory **814** is made up of detachable memory such as a SD card or the like. Note that the large-capacity nonvolatile memory **814** may be made up of non-unmountable memory, like the large-capacity nonvolatile memory **51** in the camera main unit **1**.

(176) The speaker **815**, the vibrating member **816**, and the LED **817** respectively emit sound, vibrate, and emit light, thereby notifying or warning the user of the state of the display device **800**.

(177) The audio processing unit **820** includes a left microphone **819L** and a right microphone **819R** for collecting external sound (analog signals), and performs processing of the collected analog signals to generate audio signals.

(178) The small power wireless unit **871** performs exchange of data with the camera main unit **1** by small power wireless.

(179) The high-speed wireless unit **872** performs exchange of data with the camera main unit **1** by high-speed wireless.

(180) The face sensor **806** (face detecting means) includes an infrared LED lighting circuit **821**, infrared LEDs **822**, an infrared light condensing lens **826**, and an infrared detection processing

device **827**.

(181) The infrared LED lighting circuit **821** is a circuit that has functions that are the same as those of the infrared LED lighting circuit **21** in FIG. 5, and performs control of lighting and turning off of the infrared LEDs **822**, and controls the infrared light **823** cast from the infrared LEDs **822** toward the user.

(182) The infrared light condensing lens **826** is a lens for collecting reflected light **825** of the infrared light **823**.

(183) The infrared detection processing device **827** has a sensor that detects the reflected light collected by the infrared light condensing lens **826**. This sensor converts the reflected light **825**, which has been collected, into sensor data, which is then handed to the display device control unit **801**.

(184) When the face sensor **806** illustrated in FIG. 1D is directed toward the user, the infrared light **823** cast from the infrared LED lighting circuit **821** is cast onto an infrared light illumination face **824** that is the entire face of the user, as illustrated in FIG. 6. Also, the infrared light **823** reflected at the infrared light illumination face **824** becomes the reflected light **825**, and is collected at the sensor in the infrared detection processing device **827** by the infrared light condensing lens **826**.

(185) An other-function unit **830** executes telephone functions, usage of other sensors, and so forth, which are functions unique to a smartphone, and that are unrelated to the present embodiment although details are not described herein.

(186) Usages of the camera main unit **1** and the display device **800** will be described below.

(187) FIG. 7A is a flowchart illustrating an overview of image-capturing and recording processing according to the present embodiment that is executed at the camera main unit **1** and the display device **800**.

(188) As supplementary description, which equipment illustrated in FIG. 4 is performing each step is indicated to the right of the steps in FIG. 7A. That is to say, steps **S100** to **S700** in FIG. 7A are executed at the camera main unit **1**, and steps **S800** to **S1000** in FIG. 7A are executed at the display device **800**.

(189) In step **S100**, upon the power switch **11** being turned on and power being turned on to the camera main unit **1**, the general control CPU **101** is activated and reads an activation program out from the built-in nonvolatile memory **102**. Thereafter, the general control CPU **101** executes preparatory operation processing for performing settings of the camera main unit **1** before image-capturing. Details of the preparatory operation processing will be described later with reference to FIG. 7B.

(190) In step **S200**, facial direction detection processing, in which the observation direction is inferred, is executed by the facial direction detecting unit **20** detecting the facial direction. Details of the facial direction detection processing will be described later with reference to FIG. 7C. This processing is executed at a predetermined framerate.

(191) In step **S300**, the recording direction and angle-of-view deciding unit **30** executes recording direction and range deciding processing. Details of the recording direction and range deciding processing will be described later with reference to FIG. 7D.

(192) In step **S400**, the shooting unit **40** performs image-capturing, and generates image-capturing data.

(193) In step **S500**, the image clipping and developing processing unit **50** executes recording range developing processing, in which the image-capturing data generated in step **S400** is subjected to clipping of the picture and developing processing of that range, using recording direction and angle of view information decided in step **S300**. Details of the recording range developing processing will be described later with reference to FIG. 7E.

(194) In step **S600**, the primary recording unit **60** (picture recording means) executes primary recording processing, in which the picture developed in step **S500** is saved in the primary memory **103** as picture data. Details of the primary recording processing will be described later with

reference to FIG. 14.

(195) In step **S700**, the transmission unit **70** executes transfer processing to the display device **800**, in which the picture regarding which primary recording has been performed in step **S600** is wirelessly transmitted to the display device **800** at a specified timing. Details of the transfer processing to the display device **800** will be described later with reference to FIG. 16.

(196) Step **S800** and subsequent steps are executed at the display device **800**.

(197) In step **S800**, the display device control unit **801** executes optical correction processing, in which the picture transferred from the camera main unit **1** in step **S700** is subjected to optical correction. Details of the optical correction processing will be described later with reference to FIG. 17.

(198) In step **S900**, the display device control unit **801** performs image stabilization processing on the picture subjected to optical correction in step **S800**. Details of image stabilization processing will be described later with reference to FIG. 19.

(199) Note that the order of steps **S800** and **S900** may be inverted. That is to say, image stabilization correction of the picture may be performed first, and optical correction may be performed later.

(200) In step **S1000**, the display device control unit **801** (moving image recording means) performs secondary recording for recording the picture, of which the optical correction processing and the image stabilization processing in steps **S800** and **S900** have been completed, in the large-capacity nonvolatile memory **814**, and this processing ends.

(201) Next, the subroutines in each step described with reference to FIG. 7A will be described in detail, along with the order of processing, with reference to FIGS. 7B to 7F, and other Figures and so forth.

(202) FIG. 7B is a flowchart of a subroutine for the preparatory operation processing of step **S100** in FIG. 7A. This processing will be described below using the portions illustrated in FIGS. 2A to 2F and 5.

(203) In step **S101**, determination is made regarding whether or not the power switch **11** is on. In a case in which the power remains off, the flow stands by, and when the power goes on, the flow advances to step **S102**.

(204) In step **S102**, the mode selected by the image-capturing mode switch **12** is determined. In a case in which the results of the determination are that the mode selected by the image-capturing mode switch **12** is the moving image mode, the flow advances to step **S103**.

(205) In step **S103**, various types of settings for the moving image mode are read out from the built-in nonvolatile memory **102** and saved in the primary memory **103**, following which the flow advances to step **S104**. The various types of settings in the moving image mode here include an angle-of-view settings value *ang* (set to 90° in advance in the present embodiment), and image stabilization level specified by “strong” “medium”, “off” and so forth.

(206) In step **S104**, operations of the image-capturing driver **41** are started for the moving image mode, following which the flow exits this subroutine.

(207) In a case in which the results of the determination in step **S102** are that the mode selected by the image-capturing mode switch **12** is the still image mode, the flow advances to step **S106**.

(208) In step **S106**, various types of settings for the still image mode are read out from the built-in nonvolatile memory **102** and saved in the primary memory **103**, following which the flow advances to step **S107**. The various types of settings in the still image mode here include the angle-of-view settings value *ang* (set to 45° in advance in the present embodiment), and image stabilization level specified by “strong” “medium”, “off”, and so forth.

(209) In step **S107**, operations of the image-capturing driver **41** are started for still image mode, following which the flow exits this subroutine.

(210) In a case in which the results of the determination in step **S102** are that the mode selected by the image-capturing mode switch **12** is the Pri setting mode, the flow advances to step **S108**. The

Pri setting mode here is a mode in which settings of the image-capturing mode are made to the camera main unit **1** from external equipment, such as the display device **800** or the like, and is one of the three image-capturing modes that can be switched by the image-capturing mode switch **12**. That is to say, the Pri setting mode is a mode for custom shooting. Now, the camera main unit **1** is a small-sized wearable device, and accordingly no operation switches, settings screen, or the like, for changing detailed settings thereof, are provided on the camera main unit **1**. Rather, detailed settings of the camera main unit **1** are changed by external equipment such as the display device **800**.

(211) A case will be considered in which the user desires to perform the same moving image image-capturing consecutively, once at an angle of view of 90° , and then once at an angle of view of 110° , for example. In the normal moving image mode, the angle of view of 90° is set, and accordingly in a case of performing such image-capturing, there is a need to perform image-capturing first in the normal moving image mode, following which the moving image image-capturing is temporarily ended, a settings screen for the camera main unit **1** is brought up on the display device **800**, and operations for switching the angle of view to 110° are performed. However, in a case in which some sort of event is going on, such operations at the display device **800** are bothersome.

(212) Conversely, by setting the Pri setting mode to a mode for performing moving image image-capturing at the angle of view of 110° in advance, the mode can be instantaneously switched to moving image image-capturing at the angle of view of 110° after moving image image-capturing at the angle of view of 90° has ended, simply by sliding the image-capturing mode switch **12** to “Pri”. That is to say, the user does not need to interrupt his/her current activities and perform the above-described bothersome operations.

(213) Note that the contents that are set in the Pri setting mode are not limited to angle of view, and may include the image stabilization level that is specified by “strong” “medium”, “off”, and so forth, settings for speech recognition which are not described in the present embodiment, and so on.

(214) In step **S108**, various types of settings for the Pri setting mode are read out from the built-in nonvolatile memory **102** and saved in the primary memory **103**, following which the flow advances to step **S109**. The various types of settings in the Pri setting mode here include the angle-of-view settings value ang and the image stabilization level specified by “strong” “medium”, “off”, and so forth.

(215) In step **S109**, operations of the image-capturing driver **41** are started for Pri setting mode, following which the flow exits this subroutine.

(216) Now, various types of settings for the moving image mode read out in step **S103** will be described with reference to FIG. **13**.

(217) FIG. **13** is a diagram illustrating a menu screen for various types of settings for the moving image mode, which is displayed on the display unit **803** of the display device **800**, before performing image-capturing by the camera main unit **1**. Note that portions that are the same as those in FIG. **1D** are denoted by the same signs, and description thereof will be omitted. Note that the display unit **803** has touch-panel functions, and description will be made hereinafter that the display unit **803** functions with touch operations, including operations such as swiping and so forth.

(218) In FIG. **13**, the menu screen includes a preview screen **831**, a zoom lever **832**, a recording start/stop button **833**, a switch **834**, a remaining battery charge display **835**, a button **836**, a lever **837**, and an icon display portion **838**.

(219) The preview screen **831** enables confirmation of pictures image-captured by the camera main unit **1**, and zoom amount and angle of view can be confirmed.

(220) The zoom lever **832** is an operating portion that can perform zoom settings by shifting to the right and left. In the present embodiment, a case will be described in which four values of 45° , 90° , 110° , and 130° can be set as the angle-of-view settings value ang, but an arrangement may be made in which values other than these can be set as the angle-of-view settings value ang by the zoom lever **832**.

(221) The recording start/stop button **833** is a toggle switch that has the functions of both the start switch **14** and the stop switch **15**.

(222) The switch **834** is a switch for switching image stabilization between “off” and “on”.

(223) The remaining battery charge display **835** displays remaining battery charge of the camera main unit **1**.

(224) The button **836** is a button to transition to another mode.

(225) The lever **837** is a lever for setting the level of image stabilization. Although only “strong” and “medium” can be set as the level of image stabilization in the present embodiment, other levels of image stabilization, such as “weak” for example, may be made to be settable. Also, the level of image stabilization may be settable to be continuously variable.

(226) The icon display portion **838** displays a plurality of thumbnail icons for previewing.

(227) FIG. **7C** is a flowchart of a subroutine for facial direction detection processing of step **S200** in FIG. **7A**. Before describing the details of this processing, a detection method of the facial direction by casting infrared light will be described with reference to FIGS. **8A** to **8K**.

(228) FIG. **8A** is a diagram illustrating a picture of the user as seen from the facial direction detection window **13**.

(229) Assuming a case in which the facial direction detection window **13** has no visible light cutout filter component and is capable of sufficiently transmitting visible light, and the infrared detection processing device **27** is a visible light image-capturing device, the picture in FIG. **8A** is the same as a picture image-captured by that visible light image-capturing device.

(230) A face **204** of the user, including a neck front portion **201** above the collarbones, a chin base portion **202**, a chin tip portion **203**, and a nose, is in the picture in FIG. **8A**.

(231) FIG. **8B** is a diagram illustrating a case in which fluorescent lamps of a room show up in the picture of the user as a background, as seen from the facial direction detection window **13**.

(232) A plurality of fluorescent lamps **205** in the surroundings of the user are in the picture in FIG. **8B**. In this way, various types of backgrounds and so forth are picked up by the infrared detection processing device **27**, depending on the usage conditions, and accordingly it is difficult for the facial direction detecting unit **20** and the general control CPU **101** to cut the picture of the facial portion out from the sensor data of the infrared detection processing device **27**. Nowadays, there is technology for cutting out such pictures by using artificial intelligence or the like, but this requires the general control CPU **101** to have a high capability level, and is not appropriate for the camera main unit **1** that is portable equipment.

(233) In reality, a visible light cutout filter is included in the configuration of the facial direction detection window **13**, and hardly any visible light is transmitted. Accordingly, pictures from the infrared detection processing device **27** are not pictures such as those in FIGS. **8A** and **8B**.

(234) FIG. **8C** is a diagram illustrating a picture in a case in which the user and the fluorescent lamps in the background in FIG. **8B** are imaged by the sensor of the infrared detection processing device **27** via the facial direction detection window **13**, in a state in which the infrared LEDs **22** are not lit.

(235) In the picture in FIG. **8C**, the neck and the chin of the user are dark. Conversely, the fluorescent lamps **205** emit not only visible light but infrared light components as well, and accordingly appear somewhat brightly.

(236) FIG. **8D** is a diagram illustrating a picture in a case in which the user and the fluorescent lamps in the background in FIG. **8B** are imaged by the sensor of the infrared detection processing device **27** via the facial direction detection window **13**, in a state in which the infrared LEDs **22** are lit.

(237) In the picture in FIG. **8D**, the neck and chin of the user appear brightly, unlike in FIG. **8C**. Meanwhile, the brightness in the vicinity of the fluorescent lamps **205** and so forth is unchanged.

(238) FIG. **8E** is a diagram illustrating a differential picture computed from the pictures in FIGS. **8C** and **8D**. It can be understood that the face of the user stands out.

(239) In this way, a differential image in which the face of the user is extracted (hereinafter also referred to as “facial picture”) is obtained, by the general control CPU **101** (picture acquisition means) computing a difference between images that are imaged at the sensor of the infrared detection processing device **27** with the infrared LEDs **22** lit and unlit, respectively.

(240) The facial direction detecting unit **20** according to the present embodiment employs a method of acquiring a facial picture by extracting reflection intensities of infrared light as two-dimensional images, using the infrared detection processing device **27**. The sensor of the infrared detection processing device **27** employs the same structure as a common image-capturing device, and acquires facial images one frame at a time. Vertical synchronization signals (hereinafter referred to as “V signals”) for synchronization of these frames are generated at the infrared detection processing device **27** and are output to the general control CPU **101**.

(241) FIGS. **9A** to **9E** are timing charts showing timings of lighting and turning off the infrared LEDs **22**.

(242) FIG. **9A** shows the timing of V signals being generated at the infrared detection processing device **27**. Frame synchronization and timing of lighting and turning off the infrared LEDs **22** are realized by the V signals going to Hi.

(243) In FIG. **9A**, **t1** represents a first facial image acquisition period, and **t2** represents a second facial image acquisition period. In FIGS. **9A** to **9D**, temporal axes on the horizontal axis are matched with each other.

(244) In FIG. **9B**, an H position of image signals output from the sensor of the infrared detection processing device **27** is represented on the vertical axis. As illustrated in FIG. **9B**, the infrared detection processing device **27** controls actions of the sensor thereof such that the H positions of the image signals are synchronized with the V signals. Note that since the sensor of the infrared detection processing device **27** employs the same structure as a common image-capturing device, as described above, the actions thereof are known, and accordingly detailed description of control will be omitted.

(245) FIG. **9C** illustrates a timing of switching between Hi and Low of IR-on signals output from the general control CPU **101** to the infrared LED lighting circuit **21**. The general control CPU **101** controls the switching between Hi and Low of IR-on signals so as to be synchronized with the V signals, as shown in FIG. **9C**. Specifically, the general control CPU **101** outputs a Low IR-on signal to the infrared LED lighting circuit **21** during the period at **t1**, and outputs a Hi IR-on signal to the infrared LED lighting circuit **21** during the period at **t2**.

(246) Now, during the Hi period of the IR-on signal, the infrared LED lighting circuit **21** lights the infrared LEDs **22** and the infrared light **23** is cast on the user. Conversely, during the Low period of the IR-on signal, the infrared LED lighting circuit **21** turns off the infrared LEDs **22**.

(247) FIG. **9D** shows image-capturing data output from the sensor of the infrared detection processing device **27** to the general control CPU **101**. The vertical direction represents signal intensity, indicating the quantity of reflected light **25** received. That is to say, during the period of **t1**, the infrared LEDs **22** are unlit, and accordingly there is no reflected light **25** from the facial portion of the user in this state, and image-capturing data such as shown in FIG. **8C** is acquired. Meanwhile, during the period of **t2**, the infrared LEDs **22** are lit, and accordingly there is reflected light **25** from the facial portion of the user in this state, and image-capturing data such as shown in FIG. **8D** is acquired. Accordingly, the signal intensity during the period of **t2** is higher than the signal intensity during the period of **t1** by an amount equivalent to the reflected light **25** from the facial portion of the user, as illustrated in FIG. **9D**.

(248) FIG. **9E** is the difference between the image-capturing data during each of the periods **t1** and **t2** in FIG. **9D**, and image-capturing data in which only the component of the reflected light **25** from the face of the user is extracted, such as in FIG. **8E**, is obtained.

(249) FIG. **7C** shows the facial direction detection processing in step **S200**, including the operations described above with reference to FIGS. **8C** to **8E** and FIGS. **9A** to **9E**.

(250) First, in step **S201**, when V signals output from the infrared detection processing device **27** reach a timing **V1** at which the period of **t1** starts, the flow advances to step **S202**.

(251) Next, in step **S202**, the IR-on signal is set to Low, and is output to the infrared LED lighting circuit **21**. This turns the infrared LEDs **22** off.

(252) In step **S203**, one frame worth of image-capturing data output from the infrared detection processing device **27** during the period of **t1** is read out, and this data is temporarily saved in the primary memory **103** as **Frame1**.

(253) In step **S204**, when V signals output from the infrared detection processing device **27** reach a timing **V2** at which the period of **t2** starts, the flow advances to step **S205**.

(254) In step **S205**, the IR-on signal is set to Hi, and is output to the infrared LED lighting circuit **21**. This lights the infrared LEDs **22**.

(255) In step **S206**, one frame worth of image-capturing data output from the infrared detection processing device **27** during the period of **t2** is read out, and this data is temporarily saved in the primary memory **103** as **Frame2**.

(256) In step **S207**, the IR-on signal is set to Low, and is output to the infrared LED lighting circuit **21**. This turns the infrared LEDs **22** off.

(257) In step **S208**, **Frame1** and **Frame2** are read out from the primary memory **103**, and light intensity **Fn** of the reflected light **25** of the user in FIG. **9E**, which is the difference obtained by subtracting **Frame1** from **Frame2**, is computed (this processing is generally known as “dark subtraction processing”).

(258) In step **S209**, a neck base portion position (center of neck turning) is extracted from the light intensity **Fn**.

(259) First, the general control CPU **101** (dividing means) divides the facial picture into a plurality of distance areas, on the basis of the light intensity **Fn**, which will be described with reference to FIG. **8F**.

(260) FIG. **8F** is a diagram illustrating a case in which shades of the differential picture in FIG. **8E** have been adjusted in scale, in accordance with light intensity of the reflected light **25** from the infrared light **23** cast on the face and neck portions of the user, in order to see the light quantity distribution with regard to the face and neck portions of the user.

(261) CHART **1** in FIG. **8F** is a diagram that indicates distribution of light intensity of the reflected light **25** in the facial picture in FIG. **8E**, sectioned into regions by stages of gray, for the sake of simplification of description. An **Xf** axis, used for description, passes in a direction through the middle portion of the neck of the user and the chin tip portion.

(262) In CHART **2** in FIG. **8F**, the horizontal axis indicates the light intensity on the **Xf** axis in CHART **1** in FIG. **8F**, and the vertical axis indicates the **Xf** axis. The farther toward the right on the horizontal axis, the stronger the light intensity indicated is.

(263) In CHART **1** in FIG. **8F**, the facial picture is divided into six regions (distance areas) **211** to **216** in accordance with light intensity.

(264) Region **211** is a region in which the light intensity is the strongest, and is indicated by white in the stages of gray.

(265) Region **212** is a region in which the light intensity is only slightly weaker than in region **211**, and is indicated by a gray that is considerably light in the stages of gray.

(266) Region **213** is a region in which the light intensity is even weaker than in region **212**, and is indicated by a gray that is light in the stages of gray.

(267) Region **214** is a region in which the light intensity is even weaker than in region **213**, and is indicated by a gray that is intermediate in the stages of gray.

(268) Region **215** is a region in which the light intensity is even weaker than in region **214**, and is indicated by a gray that is somewhat dark in the stages of gray.

(269) Region **216** is a region in which the light intensity is weakest, and is indicated by a gray that is darkest in the stages of gray. Upward from region **216** is black, with no light intensity.

(270) This light intensity will be described in detail below with reference to FIGS. 10A to 10D.

(271) FIGS. 10A to 10D are diagrams for describing movement of the face of the user in the up-down direction, and illustrate a state of observing the user from the left side.

(272) FIG. 10A is a diagram illustrating the user facing the front. The shooting and detecting unit 10 is situated forward of the collarbones of the user. Also, the infrared light 23 of the infrared LEDs 22 is being cast from the facial direction detection window 13 at the top of the shooting and detecting unit 10 to a lower portion of the head of the user. With Dn representing the distance from the facial direction detection window 13 to a neck base portion 200 above the collarbones of the user, db representing the distance from the facial direction detection window 13 to the chin base portion 202, and Dc representing the distance from the facial direction detection window 13 to the chin tip portion 203, it can be understood that the distance becomes longer in the order of Dn, db, and Dc. The light intensity is in inverse proportion to the distance squared, and accordingly, the light intensity when the reflected light 25 from the infrared light illumination face 24 is imaged at the sensor of the infrared detection processing device 27 becomes weaker in the order of the neck base portion 200, the chin base portion 202, and the chin tip portion 203. It can also be understood that the light intensity is even more dark with respect to the face 204 including the nose, which is at a position even farther in distance from the facial direction detection window 13 than Dc. That is to say, it can be understood that in a case such as illustrated in FIG. 10A, a picture with a distribution of light intensity such as illustrated in FIG. 8F will be acquired.

(273) Note that the configuration of the facial direction detecting unit 20 is not limited to the configuration indicated in the present embodiment, as long as the facial direction of the user can be detected. For example, an arrangement may be made in which an infrared light pattern is illuminated by the infrared LEDs 22 (infrared light pattern illumination means), and the infrared light pattern reflected from an illumination object is detected by the sensor (infrared light pattern detecting means) of the infrared detection processing device 27. In this case, the sensor of the infrared detection processing device 27 preferably is a structured light sensor. Also, the sensor of the infrared detection processing device 27 may be a sensor (infrared light phase comparing means) that performs phase comparison between the infrared light 23 and the reflected light 25, such as for example, a time of flight (ToF) sensor.

(274) Next, extraction of the position of the neck base portion in step S209 of FIG. 7C will be described with reference to FIG. 8G.

(275) CHART 1 in FIG. 8G is a diagram in which signs denoting portions of the body of the user in FIG. 10A, and double circle and black dot symbols, respectively indicating the position of the neck base portion and the position of the chin tip portion, are overlaid on FIG. 8F.

(276) The white region 211 corresponds to the neck base portion 200 (FIG. 10A), the considerably light gray region 212 corresponds to the neck front portion 201 (FIG. 10A), and the light gray region 213 corresponds to the chin base portion 202 (FIG. 10A). Also, the intermediate gray region 214 corresponds to the chin tip portion 203 (FIG. 10A), and the somewhat dark gray region 215 corresponds to the face lower portion, which is the lips at the lower portion of the face 204 (FIG. 10A) and the vicinity thereof. Further, the darkest gray region 216 corresponds to the face upper portion, which is the nose at the middle portion of the face 204 (FIG. 10A) and the vicinity thereof.

(277) As illustrated in FIG. 10A, the difference between the distances db and Dc is small as compared to distances from the facial direction detection window 13 to other portions of the user, and accordingly there is little difference in reflected light intensity between the light gray region 213 and the intermediate gray region 214.

(278) Conversely, out of distances from the facial direction detection window 13 to the portions of the user, the distance Dn is the closest distance as illustrated in FIG. 10A, and accordingly the white region 211 that corresponds to the neck base portion 200 is the portion where the reflection intensity is the strongest.

(279) Accordingly, the region 211 is the vicinity of the neck base portion 200, and a position 206

indicated by the double circle in CHART 1 in FIG. 8G, at the lateral middle in the region 211 and closest to the shooting and detecting unit 10, is set to the position of the center of neck turning (hereinafter referred to as “neck base portion position 206”) by the general control CPU 101. The processing so far is what is performed in step S209 in FIG. 7C.

(280) Next, extraction of the chin tip portion in step S210 of FIG. 7C will be described with reference to FIG. 8G.

(281) In CHART 1 in FIG. 8G, the intermediate gray region 214 that is lighter than the region 215 corresponding to the face lower portion of the face 204 including the lips is the region including the chin tip portion. It can be understood from CHART 2 in FIG. 8G that the light intensity rapidly weakens at the region 215 adjacent to the region 214, and the change in distance from the facial direction detection window 13 is great. The general control CPU 101 distinguishes the region 214 on the near side from the region 215 where the light intensity rapidly weakens as being the chin tip portion region. Further, the general control CPU 101 calculates (extracts) a position that is at the lateral center of the region 214 and that is farthest from the neck base portion position 206 (the position indicated by the black dot in CHART 1 in FIG. 8G) as a chin tip portion position 207.

(282) For example, FIGS. 8H and 8I illustrate the change when the face is directed in the right direction.

(283) FIG. 8H is a diagram illustrating a differential picture computed by the same method as in FIG. 8E, when the face of the user is directed in the right direction. FIG. 8I is a diagram in which the double circle and black dot symbols, respectively indicating the neck base portion position 206 that is the center position of neck movement and a chin tip portion position 207_r, are overlaid on FIG. 8H.

(284) The user has turned to face the right, and accordingly the region 214 moves to a region 214_r illustrated in FIG. 8I, which is to the left direction as viewed upward from the shooting and detecting unit 10 side. The region 215 that corresponds to the face lower portion including the lips in the face 204 also moves to a region 215_r, which is to the left direction as viewed upward from the shooting and detecting unit 10 side.

(285) Accordingly, the general control CPU 101 distinguishes the region 214_r on the near side from the region 215_r where the light intensity rapidly weakens as being the chin tip portion region. Further, the general control CPU 101 calculates (extracts) the position that is at the lateral center of the region 214_r and that is farthest from the neck base portion position 206 (the position indicated by the black dot in FIG. 8I) as the chin tip portion position 207_r.

(286) Thereafter, the general control CPU 101 finds a movement angle θ_r indicating how far the chin tip portion position 207_r in FIG. 8I has moved in the right direction from the chin tip portion position 207 in CHART 1 in FIG. 8G, turning centered on the neck base portion position 206. The movement angle θ_r is the angle of the face of the user in the lateral direction, as illustrated in FIG. 8I.

(287) In step S210, the infrared detection processing device 27 of the facial direction detecting unit 20 (three-dimensional detecting sensor) detects the chin tip portion position and the lateral direction angle of the face of the user by the above method.

(288) Next, detection of the face facing in the upward direction will be described.

(289) FIG. 10B is a diagram illustrating the face of the user directed in the horizontal direction, and FIG. 10C is a diagram illustrating the face of the user directed 33° upward from the horizontal direction.

(290) In FIG. 10B, the distance from the facial direction detection window 13 to the chin tip portion 203 is denoted by F_{fh}, and in FIG. 10C, the distance from the facial direction detection window 13 to a chin tip portion 203_u is denoted by F_{fu}.

(291) As illustrated in FIG. 10C, the chin tip portion 203_u moves upward along with the face, and it can be understood that F_{fu} is a longer distance than F_{fh}.

(292) FIG. 8J is a diagram illustrating a picture of the user as seen from the facial direction

detection window **13** when the face of the user is directed 33° upward from the horizontal direction. As illustrated in FIG. **10C**, since the user is facing upward, the face **204** including the lips and nose is not visible from the facial direction detection window **13**, and only as far as the chin tip portion **203** is visible. FIG. **8K** illustrates the distribution of light intensity of the reflected light **25** when the user is illuminated by the infrared light **23** at this time. FIG. **8K** is a diagram illustrating a picture of the user in which the double circle and black dot symbols, respectively indicating the neck base portion position **206** and a chin tip portion position **207_u** are overlaid on a differential picture computed by the same method as in FIG. **8E**.

(293) Six regions **211_u** to **216_u** corresponding to the light intensity in FIG. **8K** are regions that are the same as the light intensity regions as in FIG. **8F**, with “u” appended thereto. The light intensity of the chin tip portion **203** of the user is the intermediate gray region **214** in FIG. **8F**, it can be understood from FIG. **8K** that the light intensity of the chin tip portion **203** shifts to the gray side and is in the somewhat dark gray region **215_u**. In this way, as a result of F_{fu} being a longer distance than F_{fh} as illustrated in FIG. **10C**, the infrared detection processing device **27** can detect that the light intensity of the reflected light **25** from the chin tip portion **203** of the user has weakened in inverse proportion to the distance squared.

(294) Next, detection of the face facing in the downward direction will be described.

(295) FIG. **10D** is a diagram illustrating the face of the user directed 22° downward from the horizontal direction.

(296) In FIG. **10D**, a distance from the facial direction detection window **13** to a chin tip portion **203_d** is denoted by F_{fd} .

(297) As illustrated in FIG. **10D**, the chin tip portion **203_d** also moves downward along with the face, and it can be understood that F_{fd} is a shorter distance than F_{fh} , and that the light intensity of the reflected light **25** from the chin tip portion **203** is stronger.

(298) Returning to FIG. **7C**, in step **S211**, the general control CPU **101** (distance calculating means) calculates the distance from the chin tip portion to the facial direction detection window **13** from the light intensity at the chin tip portion position, detected by the infrared detection processing device **27** of the facial direction detecting unit **20** (three-dimensional detecting sensor). The angle of the face in the up-down direction is also calculated on the basis thereof.

(299) In step **S212**, the angles in the lateral direction (first detection direction) of the face, and in the up-down direction (second detection direction) perpendicular thereto, which are respectively acquired in steps **S210** and **S211**, are stored in the primary memory **103** as a three-dimensional observation direction v_i of the user (i is an optional sign). For example, an observation direction v_o in a case in which the user is observing a center portion to the front is vector information of $[0^\circ, 0^\circ]$, since a lateral direction θ_h is 0° and an up-down direction θ_v is 0° . Also, an observation direction v_r in a case in which the user is observing 45° to the right is vector information of $[45^\circ, 0^\circ]$.

(300) Note that in step **S211**, the angle of the face in the up-down direction is calculated by detecting the distance from the facial direction detection window **13**, but this method is not restrictive. For example, angular change may be calculated by comparing the level in change of light intensity of the chin tip portion **203**. That is to say, angular change of the chin portion may be calculated on the basis of gradient change of a gradient CD_u of reflected light intensity from the chin base portion **202** to the chin tip portion **203** in CHART **3** in FIG. **8K**, with respect to a gradient CD_h of reflected light intensity from the chin base portion **202** to the chin tip portion **203** in CHART **2** in FIG. **8G**.

(301) FIG. **7D** is a flowchart of a subroutine for recording direction and range deciding processing of step **S300** in FIG. **7A**. Before describing this processing in detail, an ultrawide angle picture that is the object of decision regarding recording direction and recording range in the present embodiment will be described with reference to FIG. **11A**.

(302) The camera main unit **1** according to the present embodiment achieves obtaining a picture of the observation direction by the shooting unit **40** performing image-capturing of an ultrawide angle

picture of the vicinity of the shooting and detecting unit **10** using the image-capturing lens **16** that is an ultrawide angle lens, and performing clipping of a part thereof.

(303) FIG. **11A** is a diagram illustrating an aiming field of view **125** in an ultrawide angle picture that is image-captured by the shooting unit **40**, in a case in which the user is facing toward the front.

(304) As illustrated in FIG. **11A**, a pixel region **121** of the solid-state image-capturing device **42** that is capable of image-capturing is a rectangular region. Also, an effective projection portion **122** (predetermined region) is a region in which a circular hemispherical picture that is fisheye-projected on the solid-state image-capturing device **42** by the image-capturing lens **16**, is displayed. Note that the image-capturing lens **16** is adjusted such that the center of the pixel region **121** and the center of the effective projection portion **122** agree.

(305) The outermost perimeter of the effective projection portion **122**, which is circular in form, indicates the position of 180° in field-of-view (FoV) angle. In a case in which the user is looking at the center in the horizontal and vertical directions, the aiming field of view **125**, which is the region that is image-captured and recorded, is 90°, which is half the angle from the center of the effective projection portion **122**. Note that light further on an outer side of the effective projection portion **122** can be introduced to the image-capturing lens **16** according to the present embodiment, and light to around a maximum of around 192° in FoV angle can be fisheye-projected onto the solid-state image-capturing device **42**. Note however, that when the effective projection portion **122** is exceeded, optical performance markedly deteriorates, such as extreme deterioration in resolution, loss in light quantity, increased distortion, and so forth. Accordingly, an example of clipping only within the picture projected on the pixel region **121** (hereinafter referred to simply as “ultrawide angle picture”) out of the hemispherical image displayed in the effective projection portion **122** to obtain a picture in the observation direction as a recording region will be described in the present embodiment.

(306) Note that the size of the effective projection portion **122** in the up-down direction is larger than the size of the short sides of the pixel region **121**, and accordingly the picture at the upper and lower ends of the effective projection portion **122** extends outside from the pixel region **121** in the present embodiment, but this is not restrictive. For example, a design may be made in which the configuration of the image-capturing lens **16** is changed such that the entire effective projection portion **122** fits within the pixel region **121**.

(307) Ineffective pixel regions **123** are pixel regions of the pixel region **121** that are not included in the effective projection portion **122**.

(308) The aiming field of view **125** is a region indicating a range for clipping the ultrawide angle picture to obtain the picture in the observation direction of the user, and is defined by lateral and up-down angle of view (45°, FoV angle 90° here) set in advance, centered on the observation direction. In the example in FIG. **11A**, the user is facing the front, and accordingly the center of the aiming field of view **125** is the observation direction vo that is the center of the effective projection portion **122**.

(309) There are a subject A **131** that is a child, a subject B **132** that is stairs the subject A that is a child is going to climb, and a subject C **133** that is an outdoor playset shaped like a steam locomotive, in the ultrawide angle picture illustrated in FIG. **11A**.

(310) Next, the recording direction and range deciding processing in step **S300**, which is executed to obtain the picture in the observation direction from the ultrawide angle picture described with reference to FIG. **11A** above, is shown in FIG. **7D**. This processing will be described with reference to FIGS. **12A** to **12G**, which are specific examples of the aiming field of view **125**.

(311) In step **S301**, the angle-of-view settings value ang set in advances is acquired by being read out from the primary memory **103**.

(312) In the present embodiment, all angles of view that can be used by the image clipping and developing processing unit **50** for clipping the ultrawide angle image to obtain the picture in the

observation direction, which are 45°, 90°, 110°, and 130°, are saved as angle-of-view settings values **ang** in the built-in nonvolatile memory **102**. Also, one of the angle-of-view settings values **ang** saved in the built-in nonvolatile memory **102** is set in one of steps **S103**, **S106**, and **S108**, and is saved in the primary memory **103**.

(313) Also, in step **S301**, the observation direction **vi** decided in step **S212** is decided as the recording direction, and a picture of the aiming field of view **125**, obtained by clipping the ultrawide angle image by the angle-of-view settings value **ang** acquired above, centered on this observation direction **vi**, is saved in the primary memory **103**.

(314) For example, in a case in which the angle-of-view settings value **ang** is 90° and the observation direction **vo** (vector information [0°,0°]) is detected in the facial direction detection processing (FIG. 7C), a range of 45° right and left, and 45° up and down, with a center **O** of the effective projection portion **122** as the center thereof, is set as the aiming field of view **125** (FIG. 11A). That is to say, the general control CPU **101** (relative position setting means) sets the angle of the facial direction detected by the facial direction detecting unit **20** to the observation direction **vi** that is vector information indicating the relative position as to the ultrawide angle picture.

(315) Now, in the case of observation direction **vo**, the effects of optical distortion by the image-capturing lens **16** are negligible, and accordingly the shape of the aiming field of view **125** that is set is the shape of an aiming field of view **125o** (FIG. 12A) following later-described distortion transform in step **S303**, with no change. Hereinafter in a case of observation direction **vi**, the aiming field of view **125** following distortion transform will be referred to as “aiming field of view **125i**”.

(316) Next, in step **S302**, the image stabilization level set in advance is acquired by reading out from the primary memory **103**.

(317) In the present embodiment, the image stabilization level is set in one of the steps **S103**, **S106**, and **S108**, as described above, and saved in the primary memory **103**.

(318) Also, in step **S302**, an image-stabilization reserve pixel count **Pis** is set on the basis of the image stabilization level acquired above.

(319) In image stabilization processing, the amount of blurring of the shooting and detecting unit **10** is tracked, and a picture tracking the picture in the opposite direction from the blurring direction is acquired. Accordingly, in the present embodiment, a reserve region necessary for image stabilization is provided on a perimeter of the aiming field of view **125i**.

(320) Also, in the present embodiment, a table for holding the values of the image-stabilization reserve pixel count **Pis** associated with each of the image stabilization levels is stored in the built-in nonvolatile memory **102**. For example, in a case in which the image stabilization level is “medium”, a reserve pixel region of 100 pixels, which is the image-stabilization reserve pixel count **Pis** read out from the above-described table, is set as the reserve region.

(321) FIG. 12E is a diagram illustrating an example of imparting a reserve region to the perimeter of the aiming field of view **125o** illustrated in FIG. 12A. Description will be made here regarding a case in which the image stabilization level is “medium”, i.e., the image-stabilization reserve pixel count **Pis** is 100 pixels.

(322) A dotted line portion illustrated in FIG. 12E, which is the 100 pixels of leeway (reserve region), i.e., the image-stabilization reserve pixel count **Pis** imparted to each of the top and bottom, and the right and left, of the aiming field of view **125o**, is an image-stabilization reserve pixel frame **126o**.

(323) Description made with reference to FIGS. 12A and 12E regarding a case in which the observation direction **vi** agrees with the center **O** of the effective projection portion **122** (center of the optical axis of the image-capturing lens **16**), for the sake of simplification of description. However, in a case in which the observation direction **vi** is at a peripheral portion of the effective projection portion **122**, there will be effects of optical distortion, and accordingly transform needs to be performed, such as described in the following steps.

(324) In step S303, the shape of the aiming field of view **125** set in step S301 is corrected (distortion transform) taking into consideration the observation direction v_i and the optical characteristics of the image-capturing lens **16**, thereby generating an aiming field of view **125_i**. In the same way, the image-stabilization reserve pixel count P_{is} set in step S302 is also corrected taking into consideration the observation direction v_i and the optical characteristics of the image-capturing lens **16**.

(325) For example, a case will be assumed in which the angle-of-view settings value ang is 90° , and the user is observing 45° to the right from the center o . In this case, the observation direction v_i decided in step S212 is observation direction v_r (vector information $[45^\circ, 0^\circ]$), and with the observation direction v_r as the center, a range 45° to the right and left, and 45° up and down, is the aiming field of view **125**. However, the aiming field of view **125** is corrected to an aiming field of view **125_r** illustrated in FIG. 12B, taking into consideration the optical characteristics of the image-capturing lens **16**.

(326) As illustrated in FIG. 12B, the aiming field of view **125_r** becomes wider the closer to the peripheral portion of the effective projection portion **122**, and also, the position of the observation direction v_r is slightly toward an inner side from the center of the aiming field of view **125_r**. This is because an optical design close to that of a stereographic projection fisheye lens is made for the image-capturing lens **16** in the present embodiment. Note that this relation changes when the image-capturing lens **16** is designed as an equidistance projection fisheye lens, an equisolid angle projection fisheye lens, an orthographic projection fisheye lens, or the like, and accordingly correction matching the optical characteristics thereof is performed on the aiming field of view **125**.

(327) FIG. 12F is a diagram illustrating an example of imparting a reserve region of image stabilization level “medium”, which is the same image stabilization level as the reserve region in FIG. 12E, at a perimeter of the aiming field of view **125_r** illustrated in FIG. 12B.

(328) In the image-stabilization reserve pixel frame **126_o** (FIG. 12E), the 100 pixels of leeway that is the image-stabilization reserve pixel count P_{is} is imparted to each of the top and bottom, and the right and left, of the aiming field of view **125_o**. Conversely, in an image-stabilization reserve pixel frame **126_r** (FIG. 12F), the image-stabilization reserve pixel count P_{is} is corrected, and increases the closer to the peripheral portion of the effective projection portion **122**.

(329) In this way, the shape of the reserve region necessary for image stabilization that is provided on the perimeter of the aiming field of view **125_r** is also subjected to a greater correction amount the closer to the peripheral portion of the effective projection portion **122**, in the same way as with the shape of the aiming field of view **125_r**, as indicated by the image-stabilization reserve pixel frame **126_r** in FIG. 12F. This also is because optical design close to that of a stereographic projection fisheye lens is made for the image-capturing lens **16** in the present embodiment. Note that this relation changes when the image-capturing lens **16** is designed as an equidistance projection fisheye lens, an equisolid angle projection fisheye lens, an orthographic projection fisheye lens, or the like, and accordingly correction that matches the optical characteristics thereof is performed on the image-stabilization reserve pixel frame **126_r**.

(330) The processing of sequentially switching the shape of the aiming field of view **125** and the reserve region, taking into consideration the optical characteristics of the image-capturing lens **16**, which is executed in step S303, is complicated processing. Accordingly, in the present embodiment, the processing of step S303 is executed using a table that holds the aiming field of view **125_i** for each observation direction v_i , and the shape of the reserve region thereof, which is in the built-in nonvolatile memory **102**. Note that depending on the optical design of the image-capturing lens **16** mentioned earlier, a computation expression may be kept in the general control CPU **101**, with optical distortion values being calculated by this computation expression.

(331) In step S304, the position and the size of a picture recording frame are calculated.

(332) As described above, in step S303, a reserve region necessary for image stabilization is

provided on the perimeter of the aiming field of view **125i**, and is calculated as the image-stabilization reserve pixel frame **126i**. However, depending on the position of the observation direction v_i , the shape becomes quite unusual, like the image-stabilization reserve pixel frame **126r**, for example.

(333) The general control CPU **101** is capable of performing developing processing on just a range of such an unusual shape, and clipping the picture. However, using a picture that is not rectangular when recording as picture data in step **S600** or transferring to the display device **800** in step **S700** is not commonly performed. Accordingly, in step **S304**, the position and the size of a picture recording frame **127i** that is rectangular in shape, which envelops the entirety of this image-stabilization reserve pixel frame **126i**, are calculated.

(334) In FIG. **12F**, a picture recording frame **127r**, indicated by the chain line calculated in step **S304** with respect to the image-stabilization reserve pixel frame **126r**, is illustrated.

(335) In step **S305**, the position and the size of the picture recording frame **127i** calculated in step **S304** are recorded in the primary memory **103**.

(336) In the present embodiment, coordinates X_i , Y_i of the upper left of the picture recording frame **127i** in the ultrawide angle picture are recorded as the position of the picture recording frame **127i**, and a lateral width W_{Xi} and a vertical width W_{Yi} of the picture recording frame **127i** from the coordinates X_i , Y_i are recorded as the size of the picture recording frame **127i**. For example, coordinates X_r , Y_r and lateral width W_{Xr} and vertical width W_{Yr} illustrated in FIG. **12F** are recorded in step **S305** with respect to the picture recording frame **127r** illustrated in FIG. **12F**. Note that the coordinates X_i , Y_i are XY coordinates with a predetermined reference point, specifically, the optical center of the image-capturing lens **16**, as the point of origin.

(337) Once the image-stabilization reserve pixel frame **126i** and the picture recording frame **127i** are decided in this way, the flow exits the subroutine for step **S300** shown in FIG. **7D**.

(338) Note that in the description so far, description has been made using observation directions including horizontal 0° , i.e., observation direction v_o (vector information $[0^\circ, 0^\circ]$) and observation direction v_r (vector information $[45^\circ, 0^\circ]$) as examples of the observation direction v_i , for the sake of simplification of the complicated description of optical distortion transform. However, in reality, the observation direction v_i of the user will be in various directions. Accordingly, the recording range developing processing executed in such cases will be described below.

(339) For example, an aiming field of view **125l** in a case of an angle-of-view settings value ang of 90° and an observation direction v_1 $[-42^\circ, -40^\circ]$ will be as illustrated in FIG. **12C**.

(340) Also, even when the observation direction v_1 (vector information $[-42^\circ, -40^\circ]$) is the same as the aiming field of view **125l**, in a case in which the angle-of-view settings value ang is 45° , the aiming field of view will be an aiming field of view **1281** that is one size smaller than the aiming field of view **125l**, as illustrated in FIG. **12D**. Further, an image stabilization reserve pixel frame **1291** and a picture recording frame **1301**, such as illustrated in FIG. **12G**, are set for the aiming field of view **1281**.

(341) Step **S400** is basic operations of image-capturing, and since a common sequence of the shooting unit **40** is used, details will be left to other literature and description will be omitted here. Note that in the preset embodiment, the image-capturing signal processing circuit **43** in the shooting unit **40** also performs processing of converting signals in unique output formats (examples of such standards include Mobile Industry Processor Interface (MIPI) and Scalable Low Voltage Signaling (SLVS)) that are output from the solid-state image-capturing device **42** into image-capturing data of a common sensor readout format.

(342) Note that in a case in which the mode selected by the image-capturing mode switch **12** is the moving image mode, the shooting unit **40** starts recording in accordance with the start switch **14** being pressed. Thereafter, the recording ends when the stop switch **15** is pressed. Conversely, in a case in which the mode selected by the image-capturing mode switch **12** is the still image mode, the shooting unit **40** performs image-capturing of a still image each time the start switch **14** is

pressed.

(343) FIG. 7E is a flowchart of a subroutine for recording range developing processing of step S500 in FIG. 7A.

(344) In step S501, raw data is acquired for the entire region of the image-capturing data (ultrawide angle picture) generated by the shooting unit 40 in step S400, and is input to a picture intake portion of the general control CPU 101, called a head portion, which is omitted from illustration.

(345) Next, in step S502, clipping is performed to obtain the portion of the picture recording frame 127i from the ultrawide angle picture acquired in step S501 on the basis of the coordinates Xi, Yi and the lateral width WXi and vertical width WYi recorded in the primary memory 103 in step S305. After this clipping, crop developing processing (FIG. 7F) that is made up of the steps S503 to S508 executed thereafter is started with respect to only pixels in the image-stabilization reserve pixel frame 126i. This enables the computation amount to be markedly reduced as compared to a case of performing developing processing on the entire region of the ultrawide angle picture read in step S501, and computing time and electric power consumption can be reduced.

(346) Note that in a case in which the mode selected by the image-capturing mode switch 12 is the moving image mode, the processing of steps S200 and S300, and the processing of step S400, are executed in parallel as illustrated in FIG. 7F, at the same framerate as each other or different framerates from each other. That is to say, each time that the entire region of one frame worth of raw data generated by the shooting unit 40 is acquired, crop developing processing is performed on the basis of the coordinates Xi, Yi and the lateral width WXi and vertical width WYi recorded in the primary memory 103 at that point in time.

(347) Upon the crop developing processing being started with respect to pixels in the image-stabilization reserve pixel frame 126i, first, color complementation is performed in step S503 to complement color pixel information arrayed according to a Bayer filter array.

(348) Following white balance adjustment being performed in step S504, color conversion is performed in step S505.

(349) In step S506, gamma correction is performed, in which gradients are corrected in accordance with a gamma correction value that is set in advance.

(350) In step S507, edge enhancement is performed in accordance with image size.

(351) In step S508, compression and other processing are performed, thereby converting into a data format capable of primary saving, and following recording thereof in the primary memory 103, the flow exits this subroutine. Details of this data format capable of primary saving will be described later.

(352) Note that the order and presence/absence of processing in the crop developing processing executed in steps S503 to S508 can be performed in accordance with the camera system, and do not limit the present invention.

(353) Also, in a case in which the moving image mode is selected, the processing of steps S200 to S500 is repeatedly executed until recording ends.

(354) According to the present processing, the computation amount can be markedly reduced as compared to a case of performing developing processing on the entire region read in step S501. This enables a microcontroller that is inexpensive and has low electric power consumption to be used as the general control CPU 101. Also, heat generated at the general control CPU 101 can be suppressed, and the life of the batteries 94 can be extended.

(355) Also, in the present embodiment, optical correction processing (step S800 in FIG. 7A) and image stabilization processing (step S900 in FIG. 7A) of the picture are not performed at the camera main unit 1, but rather are performed at the display device control unit 801 following transfer thereof to the display device 800, in order to lighten the control load on the general control CPU 101. Accordingly, if only data of the picture that has been partially clipped from the projected ultrawide angle picture is sent to the display device 800, optical correction processing and image stabilization processing cannot be performed. That is to say, if only data of the picture that has been

partially clipped is available, such processing cannot be correctly executed at the display device **800**, since there is no position information to be used for substituting into the expression for optical correction processing or to be referenced by the correction table at the time of image stabilization processing. Accordingly, in the present embodiment, not only the data of the picture that has been clipped, but correction data including the information of the position of clipping this picture from the ultrawide angle picture, and so forth, is also transmitted from the camera main unit **1** to the display device **800**.

(356) Now, in a case in which the picture that has been clipped is a still image picture, the data of the still image picture and the correction data are in a one-on-one relation, and accordingly even if the data of the still image picture and the correction data are separately transmitted to the display device **800**, optical correction processing and image stabilization processing can be correctly performed at the display device **800**. Conversely, in a case in which the picture that has been clipped is a moving image picture, and the data of the moving image picture and the correction data are separately transmitted to the display device **800**, judging which of the frames of the moving image picture the transmitted correction data corresponds to is difficult. In particular, if the clock rate of the general control CPU **101** in the camera main unit **1** and the clock rate of the display device control unit **801** in the display device **800** are slightly different, synchronization between the general control CPU **101** and the display device control unit **801** will be lost after a few minutes' worth of moving image image-capturing. As a result, problems occur such as the display device control unit **801** correcting a frame to be processed using correction data that is different from the correction data corresponding to the frame to be processed.

(357) Accordingly, in the present embodiment, at the time of transmitting data of a moving image picture that is clipped, from the camera main unit **1** to the display device **800**, the correction data thereof is appropriately imparted to the data of the moving image picture. A method thereof will be described below.

(358) FIG. **14** is a flowchart of a subroutine for primary recording processing of step **S600** in FIG. **7A**. This processing will be described below with reference to FIG. **15**. FIG. **14** shows processing in a case in which the mode selected by the image-capturing mode switch **12** is the moving image mode. Note that in a case in which the mode selected by the image-capturing mode switch **12** is the still image mode, this processing starts from the processing of step **S601** and ends when the processing of step **S606** ends.

(359) In step **S601a**, the general control CPU **101** reads out, from a moving image picture developed in the recording range developing processing (FIG. **7E**), an image of one frame that is unprocessed with regard to the processing of steps **S601** to **S606**. Also, the general control CPU **101** (metadata generating means) generates correction data, which is metadata of the frame that is read out.

(360) In step **S601**, the general control CPU **101** attaches information of the clipping position of the image of the frame read out in step **S600** to the correction data. The information attached here is the coordinates X_i , Y_i of the picture recording frame **127i** acquired in step **S305**. Note that the information attached here may be vector information indicating the observation direction v_i .

(361) In step **S602**, the general control CPU **101** (optical correction value acquiring means) acquire an optical correction value. The optical correction value is the optical distortion value set in step **S303**. Alternatively, the optical correction value may be a correction value in accordance with lens optical characteristics, such as a relative illuminance correction value, diffraction correction, or the like.

(362) In step **S603**, the general control CPU **101** attaches the optical correction value used for distortion transform in step **S303**, which is acquired in step **S602**, to the correction data.

(363) In step **S604**, the general control CPU **101** judges whether in the image stabilization mode or not. Specifically, in a case in which the image stabilization mode set in advance is “medium” or “strong”, judgment is made of being in the image stabilization mode, and the flow advances to step

S605. Conversely, in a case in which the image stabilization mode set in advance is “off”, judgment is made of not being in the image stabilization mode, and the flow advances to step **S606**. Note that the reason why step **S605** is skipped in a case in which the image stabilization mode is “off” is to reduce the computation data amount for the general control CPU **101** and the amount of data wirelessly transmitted, which consequently enables electric power conservation and reduced heat generation at the camera main unit **1**. While description has been made here regarding reduction of data used for image stabilization processing, data reduction may also be performed regarding relative illuminance correction value, presence/absence of diffraction correction, and so forth, included in the optical correction value acquired in step **S602**.

(364) Although the image stabilization mode is set in advance by operations performed by the user using the display device **800** in the present embodiment, this may be set as initial settings of the camera main unit **1**. Also, in a case of a camera system in which presence/absence of image stabilization is switched after transfer to the display device **800**, step **S604** is omitted, and the flow advances from step **S603** directly to step **S605**.

(365) In step **S605**, the general control CPU **101** (movement amount detecting means) attaches the image stabilization mode acquired in step **S302**, and gyro data in the moving image being image-captured associated with the frame read out in step **S600** in the primary memory **813**, to the correction data.

(366) In step **S606**, a picture file **1000** (FIG. 15) is updated by data obtained by encoding data of the image of the frame read out in step **S600**, and the correction data to which various types of data are attached in steps **S601** to **S605**. Note that in a case in which the first frame of the moving image picture is read out in step **S601a**, the picture file **1000** is generated in step **S606**.

(367) Whether reading out of images of all frames of the moving image picture developed in the recording range developing processing (FIG. 7E) has ended is distinguished in step **S607**, and if not ended, the flow returns to step **S601a**. Conversely, in a case of the reading having ended, the flow exits this subroutine. The picture file **1000** that is generated is saved in the built-in nonvolatile memory **102**. Saving may be performed to not only the primary memory **813** and the built-in nonvolatile memory **102** that are described above, but performed in the large-capacity nonvolatile memory **51** as well. Transfer processing of the picture file **1000** that is generated to the display device **800** (step **S700** in FIG. 7A) may be immediately performed, and saved in the primary memory **813** of the display device **800** after transfer thereto.

(368) Now, in the present embodiment, “encoding” refers to integrating picture data and correction data into a single file, but at this time, compression of the picture data, or compression of the integrated picture data and correction data, may be performed.

(369) FIG. 15 is a diagram illustrating a data structure of the picture file **1000**.

(370) The picture file **1000** is made up of a header **1001** and a frame **1002**. The frame **1002** is made up of frame datasets in which each image of a frame making up the moving image picture, and frame meta corresponding thereto, form a set. That is to say, there is a count of frame datasets in the frame **1002** corresponding to the total count of frames of the moving image picture.

(371) In the present embodiment, the frame meta is information in which are encoded the clipping position (in-picture position information), and correction data in which optical correction value and gyro data are attached as necessary, but is not limited to this. For example, other information may be attached to the frame meta in accordance with the image-capturing mode selected by the image-capturing mode switch **12**, or the amount of information of the frame meta may be changed by deleting information in the frame meta, or the like.

(372) An offset value to the frame dataset of each frame, or a top address, is recorded in the header **1001**. Alternatively, metadata such as time and size, corresponding to the picture file **1000**, may be saved therein.

(373) Thus, in the primary recording processing (FIG. 14), the picture file **1000** in which the frames of the moving image picture developed in the recording range developing processing (FIG.

7E), and metadata thereof, form a set, is transferred to the display device **800**. Accordingly, even if the clock rate of the general control CPU **101** in the camera main unit **1** and the clock rate of the display device control unit **801** in the display device **800** are slightly different, the display device control unit **801** can accurately execute correction processing of the moving image picture developed at the camera main unit **1**.

(374) Note that in the present embodiment, the optical correction value is included in the frame meta, but the optical correction value may be imparted to the entire picture.

(375) FIG. **16** is a flowchart of a subroutine for transfer processing to the display device **800** in step **S700** in FIG. **7A**. FIG. **16** shows processing in a case in which the mode selected by the image-capturing mode switch **12** is the moving image mode. Note that in a case in which the mode selected by the image-capturing mode switch **12** is the still image mode, this processing starts from the processing of step **S702**.

(376) In step **S701**, judgment is made regarding whether recording of the moving image picture by the shooting unit **40** (step **S400**) has ended, or recording is continuing. In a case in which the moving image picture is being recorded (during moving image image-capturing) here, this is a state in which recording range developing processing for each frame (step **S500**), and updating of the picture file **1000** (step **S606**) in the primary recording processing (step **S600**) are being sequentially performed. The electric power load of wireless transmission is great, and accordingly performing in conjunction with recording necessitates a larger battery capacity for the batteries **94**, requires separate heatproofing measures, and so forth. Also, from the perspective of computation capabilities, performing wireless transfer in conjunction with recording increases the computing load, which requires a general control CPU **101** with high specs to be prepared, raising costs. In light of the above, in the present embodiment, the flow advances to step **S702** after awaiting recording of the moving image picture to end (YES in step **S701**), and thereupon establishes a connection with the display device **800**. Note however, that in a case in which the camera system according to the preset embodiment has leeway in electric power supplied from the batteries **94**, and there is no need for separate heatproofing measures, the display device **800** may be connected thereto in advance, such as at the time of activating the camera main unit **1**, prior to starting recording, or the like.

(377) In step **S702**, a connection is established with the display device **800** via the high-speed wireless unit **72**, in order to transfer the picture file **1000** with a great data amount to the display device **800**. Note that the small power wireless unit **71** is used for transfer of low-resolution pictures (or pictures) to the display device **800** for confirmation of angle of view, or for exchange of various types of settings values with the display device **800**, but is not used for transfer of the picture file **1000** since this would take time for transferring.

(378) In step **S703**, the picture file **1000** is transferred to the display device **800** via the high-speed wireless unit **72**, and at the point of ending transfer, the flow advances to step **S704**. After closing the connection to the display device **800**, the flow exits this subroutine.

(379) Although a case has been described so far in which a single picture file containing images of all frames of a single moving image picture is transferred, a plurality of picture files segmented in time increments may be transferred in a case in which the moving image picture is long and spans several minutes. Even in a case of transferring a single moving image picture to the display device **800** as a plurality of picture files, employing the data structure illustrated in FIG. **15** for the picture file enables the moving image picture to be corrected at the display device **800** with no timing discrepancy with regard to the correction data.

(380) FIG. **17** is a flowchart of a subroutine for optical correction processing of step **S800** in FIG. **7A**. This processing will be described below with reference to FIGS. **18A** to **18F**. Note that as described above, this processing is processing that the display device control unit **801** of the display device **800** executes.

(381) In step **S801**, first, the display device control unit **801** (picture file receiving means) receives,

from the camera main unit **1**, the picture file **1000**, which was transferred to the display device **800** in the transfer processing (step **S700**). Thereafter, the display device control unit **801** (first extracting means) acquires the optical correction value extracted from the picture file **1000** that has been received.

(382) Next, in step **S802**, the display device control unit **801** (second extracting means) acquires a picture (an image of one frame obtained in the moving image image-capturing) from the picture file **1000**.

(383) In step **S803**, the display device control unit **801** (frame picture correcting means) performs optical correction of the picture acquired in step **S802**, using the optical correction value acquired in step **S801**, and saves the corrected picture in the primary memory **813**. In a case of performing clipping from the picture that is acquired in step **S802** at the time of performing optical correction, clipping is performed in a range of the image that is narrower than the developing range decided in step **S303** (aiming field of view **125i**), and processing is performed (clipping and developing region).

(384) FIGS. **18A** to **18F** are diagrams for describing a case of implementing distortion aberration correction in step **S803** of FIG. **17**.

(385) FIG. **18A** is a diagram illustrating the position of a subject **1401** as viewed by the naked eye of the user at the time of image-capturing, and FIG. **18B** is a diagram illustrating an image of the subject **1401** formed on the solid-state image-capturing device **42**.

(386) FIG. **18C** is a diagram illustrating a developing region **1402** in the image in FIG. **18B**. The developing region **1402** here is the clipping and developing region described earlier.

(387) FIG. **18D** is a diagram illustrating the clipping and developing region in which the image of the developing region **1402** has been clipped, and FIG. **18E** is a diagram illustrating a picture in which the clipping and developing region of FIG. **18D** has been subjected to distortion aberration correction. Clipping processing is performed on the clipping and developing picture at the time of performing distortion aberration correction, and accordingly, the picture illustrated in FIG. **18E** has an even smaller angle of view than the clipping and developing region illustrated in FIG. **18D**.

(388) FIG. **19** is a flowchart of a subroutine for image stabilization processing of step **S900** in FIG. **7A**. This processing will be described below with reference to FIG. **25**. Note that as described above, this processing is processing that the display device control unit **801** of the display device **800** executes.

(389) In step **S901**, the gyro data of the current frame and the preceding frame, and a blurring amount $V_{\text{sub.n.sup.Det}}$ calculated in later-described step **S902** with respect to the preceding frame, are acquired from the frame meta of the picture file **1000**. Thereafter, a general blurring amount $V_{\text{sub.n.sup.Pre}}$ is calculated from these pieces of information. Note that in the present embodiment, the current frame is the frame currently being processed, and the preceding frame is the frame one frame before the current frame.

(390) In step **S902**, a detailed blurring amount $V_{\text{sub.n.sup.Det}}$ is found from the picture. Detection of the blurring amount is performed by calculating how far feature points in the image of the current frame have moved as compared to the preceding frame.

(391) Known methods can be employed for extracting the feature points. For example, an arrangement may be made in which only luminance information is extracted from the image of the frame and a luminance information image is generated, an image obtained by performing shifting thereof by one to several pixels is subtracted from the original image, and pixels of which an absolute value thereof is no smaller than a threshold value are extracted as the feature points. Alternatively, an arrangement may be made in which an image, obtained by subjecting the above luminance information image to a high-pass filter, is subtracted from the original luminance information image, and extracted edges are extracted as the feature points.

(392) The amount of movement is calculated by calculating the difference for a plurality of number of times while shifting the luminance information images of the current frame and the preceding

frame one to several pixels at a time, and calculating positions where difference is small at pixels of feature points.

(393) A plurality of feature points are necessary, as described later, and accordingly, the images of the current frame and the preceding frame are preferably divided into a plurality of blocks to perform extraction of feature points. Although block division depends on the number of pixels and aspect ratio of the image, generally, 4×3 for 12 blocks, to 96×64 blocks, is preferable. The reason is that when the number of blocks is too small, correction of trapezoidal distortion due to the tilt angle of the shooting unit **40** of the camera main unit **1**, rotation of optical axis direction, and so forth, cannot be accurately performed, but when the number of blocks is too great, the size of each block becomes small and the feature points are close to each other, and accordingly include error. Accordingly, an optimal number of blocks is appropriately selected in accordance with pixel count, ease of finding feature points, angle of view of the subject, and so forth.

(394) Calculation of the amount of movement requires calculating difference a plurality of number of times while shifting the luminance information images of the current frame and the preceding frame one to several pixels at a time, and accordingly the amount of calculations is great. Accordingly, the amount of calculations for the actual movement amount can be markedly reduced by performing difference calculations for just the neighborhood thereof, in order to calculate just how many pixels away from the blurring amount $V_{sub.n.sup.Pre}$.

(395) Next, in step **S903**, image stabilization correction is performed using the detailed blurring amount $V_{sub.n.sup.Det}$ acquired in step **S902**, and thereafter the flow exits this subroutine.

(396) Note that Euclidean transformation that is capable of rotation and translation, affine transformation that is capable of the same, and further projective transformation that is capable of trapezoidal correction, and so forth, are conventionally known as methods for image stabilization processing.

(397) Euclidean transformation is capable of handling cases of movement and rotation on the X axis and Y axis, but blurring that actually occurs when performing image-capturing by the shooting unit **40** of the camera main unit **1** also includes camera shaking in the front and back directions, and in the panning and tilting directions. Accordingly, image stabilization correction is performed in the present embodiment using affine transformation that is capable of correcting enlargement, skewing, and so forth as well. In affine transformation, movement of a coordinates (x, y) of a reference feature point to coordinates (x', y') is expressed in the following Expression 100.

$$(398) \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (\text{Expression 100})$$

(399) Affine transformation is capable of calculating the 3×3 matrix in Expression 100 as long as displacement of at least three feature points can be detected. However, in a case in which the detected feature points are at close distance from each other, or are on a straight line, image stabilization correction at positions further away from the feature points, or positions away from the straight line become inaccurate. Accordingly, the detected feature points are preferably selected so as to be at distances far away from each other, and not on a straight line. Thus, in a case in which a plurality of feature points are detected, feature points that are close to each other are omitted, and the remainder is subjected to normalization by the method of least squares.

(400) FIG. **18F** is an image illustrating a picture obtained by subjecting the distortion-corrected picture illustrated in FIG. **18E** to the image stabilization correction of step **S903**. Clipping processing is performed at the time of image stabilization correction, and accordingly the picture illustrated in FIG. **18F** has a smaller angle of view than the picture illustrated in FIG. **18E**.

(401) Performing such image stabilization processing enables a quality picture, with blurring corrected, to be obtained.

(402) The series of operations executed at the camera main unit **1** and the display device **800**

included in the camera system according to the present embodiment have been described so far.

(403) When a user turns the power switch **11** to on, and thereafter selects the moving image mode with the image-capturing mode switch **12**, and simply performs observation to the front, without directing his/her face to the up or down, or right or left directions, the facial direction detecting unit **20** first detects the observation direction v_0 (vector information $[0^\circ, 0^\circ]$) (FIG. **12A**). Thereafter, the recording direction and angle-of-view deciding unit **30** clips the picture (FIG. **11B**) of the aiming field of view **125o** illustrated in FIG. **12A**, from the ultrawide angle picture projected on the solid-state image-capturing device **42**.

(404) Thereafter, upon the user starting to observe the child (subject A **131**) in FIG. **11A**, for example, without operating the camera main unit **1** at all, the facial direction detecting unit **20** first detects the observation direction v_1 (vector information $[-42^\circ, -40^\circ]$) (FIG. **11C**). Subsequently, the recording direction and angle-of-view deciding unit **30** clips the picture of the aiming field of view **125l** (FIG. **11C**) from the ultrawide angle picture image-captured by the shooting unit **40**.

(405) In this way, in steps **S800** and **S900**, optical correction processing and image stabilization processing are performed by the display device **800** on pictures clipped in various types of shapes in accordance with the observation direction. Thus, even in a case of clipping a picture with a great amount of distortion, such as the aiming field of view **125l** (FIG. **11C**), a picture can be obtained with distortion and blurring corrected, in which the child (subject A **131**) is at the center, such as in FIG. **11D**, even if the general control CPU **101** of the camera main unit **1** is a low-spec item. That is to say, the user can obtain pictures in which his/her observation direction is image-captured, without touching the camera main unit **1**, other than turning the power switch **11** on and selecting the mode by the image-capturing mode switch **12**.

(406) Now, the Pri setting mode will be described. As described earlier, the camera main unit **1** is a small-size wearable device, and accordingly no operating switches, settings screen, or the like for changing the detailed settings thereof are provided to the camera main unit **1**. Accordingly, detailed settings of the camera main unit **1** are changed by external equipment such as the display device **800** (the settings screen (FIG. **13**) of the display device **800** in the present embodiment).

(407) A case will be considered in which, for example, the user desires to perform the same moving image image-capturing consecutively, once at an angle of view of 90° , and then once at an angle of view of 45° . In the normal moving image mode, the angle of view of 90° is set, so in a case of performing such image-capturing, there is a need to perform image-capturing first in the normal moving image mode, following which the moving image image-capturing is temporarily ended, a settings screen for the camera main unit **1** is brought up on the display device **800**, and operations for switching the angle of view to 45° are performed. However, in a case performing continuous image-capturing, operations at the display device **800** are bothersome, and a chance to shoot a desired picture may be missed.

(408) Conversely, by setting the Pri setting mode to a mode for performing moving image image-capturing at the angle of view of 45° in advance, the mode can be instantaneously switched to zoomed-in moving image image-capturing at the angle of view of 45° after moving image image-capturing at the angle of view of 90° has ended, simply by sliding the image-capturing mode switch **12** to "Pri". That is to say, the user does not need to interrupt his/her current activities of image-capturing and perform the above-described bothersome operations.

(409) Note that the contents that are set in the Pri setting mode are not limited to changing the angle of view, and may include the image stabilization level that is specified by "strong", "medium", "off", and so forth, changing settings for speech recognition which are not described in the present embodiment, and so on.

(410) For example, when the user is in the above-described state of performing image-capturing, and switches from the moving image mode to the Pri setting mode by the image-capturing mode switch **12**, while continuing to observe the child (subject A **131**), the angle-of-view settings value ang is changed from 90° to 45° . In this case, the recording direction and angle-of-view deciding

unit **30** (setting means) clips the picture of the aiming field of view **1281** indicated by the dotted line frame in FIG. **11E**, from the ultrawide angle picture image-captured by the shooting unit **40**. (411) Optical correction processing and image stabilization processing are performed by the display device **800** in steps **S800** and **S900** in the Pri setting mode, as well. Thus, a picture can be obtained with distortion and blurring corrected, in which the child (subject A **131**) is zoomed in on at the center, such as in FIG. **11F**, even if the general control CPU **101** of the camera main unit **1** is a low-spec item. Although description has been made by way of an example in which the angle-of-view settings value *ang* is changed from 90° to 45° in the moving image mode, the same is true for the still image mode as well. Also, the same is true in a case in which the angle-of-view settings value *ang* for moving images is 90° and the angle-of-view settings value *ang* for still images is 45°, as well.

(412) Thus, the user can obtain zoom pictures image-captured in his/her observation direction, simply by switching the mode of the camera main unit **1** with the image-capturing mode switch **12**.

(413) Note that while description is made in the present embodiment regarding a configuration in which the facial direction detecting unit **20** and the shooting unit **40** are integrated in the camera main unit **1**, this is not limiting, as long as the facial direction detecting unit **20** is worn somewhere on the body of the user other than the head, and the shooting unit **40** is worn on the body of the user. For example, the shooting and detecting unit **10** according to the present embodiment may be disposed on a shoulder or the abdomen. However, in the case of a shoulder, it is conceivable that disposing the shooting unit **40** on the right shoulder would result in subjects to the left side being shielded by the head, and accordingly a configuration is preferable in which a plurality of shooting means are disposed for supplementation, at parts of the body including the left shoulder and so forth. Also, in the case of the abdomen, disparity occurs spatially between the shooting unit **40** and the head, and accordingly it is preferable to be able to perform correction calculation of the observation direction, in order to correct this disparity.

Second Embodiment

(414) A second embodiment is an embodiment that enables image-capturing of a desired picture, even when the user directs his/her line of sight (face) away from the direction of the subject while performing image-capturing of a moving image. In the second embodiment, even in a case in which the user directs his/her face toward the display device **800** displaying the state of the camera main unit **1**, for example, the camera main unit **1** can continue to perform image-capturing of a picture in the direction of view of the user before changing his/her facial direction.

(415) An image-capturing system according to the second embodiment will be described with reference to FIGS. **20** to **26**. The image-capturing system includes the camera main unit **1**, and the display device **800** that is communicable with the camera main unit **1**. Configurations of the camera main unit **1** and the display device **800** according to the second embodiment that are the same as the configurations in the first embodiment will be denoted using the same signs, and repetitive description will be omitted. Configurations of which processing is different from in the first embodiment will be described by additionally providing details.

(416) The camera main unit **1** is worn on the neck portion of the user, and accordingly the user does not see a display portion (screen, LED lamp, etc.) displaying the state of the camera main unit **1** while performing image-capturing. The user can confirm the state of the camera main unit **1** by the display device **800** being communicably connected to the camera main unit **1**, and a menu screen for various types of settings of the moving image mode, such as described with reference to FIG. **13**, being displayed. The display device **800** can exchange various types of data with the camera main unit **1** by applications or firmware for cooperation with the camera main unit **1**. The state of the camera main unit **1** includes, for example, confirmation of whether performing image-capturing or image-capturing is stopped, remaining charge of the batteries **94**, remaining capacity of the built-in nonvolatile memory **102**, and so forth.

(417) However, when the user directs his/her face toward the display device **800** while performing

image-capturing, the camera main unit **1** records a range including the display device **800** that exists in the facial direction of the user as a clipping range. Accordingly, image-capturing of the picture in the direction that the user originally desired is not performed. Further, there is a possibility that the display device **800** is displaying a screen including personal information, such as credit card information or the like, besides the screen for confirming the state of the camera main unit **1**, and there are cases in which the user does not want to leave the picture of the display device **800** recorded.

(418) In order to keep the display device **800** from being an object of image-capturing, the user is forced to hold the display device **800** at a position where it is not readily image-captured by the camera main unit **1**, and to direct his/her line of sight toward the display device **800** without changing the orientation of his/her face. Confirming the state of the camera main unit **1** displayed on the display device **800** without changing the orientation of the face during image-capturing is a tiring action for the user, and visual confirmation of information of the camera main unit **1** that the user wants to confirm is not easy.

(419) The camera main unit **1** according to the present embodiment is arranged so that a desired picture can continue to be recorded even when the user directs his/her facial direction toward the display device **800** to confirm the state of the camera main unit **1**, without interrupting image-capturing. Note that the object of the user directing his/her face toward the display device **800** is not limited to state confirmation of the camera main unit **1**, and may be for operations or the like to respond to a telephone call, an email, or the like. In the example below, description will be made assuming a flow in which the user directs his/her face toward the display device **800**, displays the state of the camera main unit **1** on the screen of the display device **800**, confirms the state of the camera main unit **1**, and returns his/her face to the original position.

(420) During a period from the user starting to change his/her facial direction toward the display device **800** up to returning his/her facial direction to the original position, the facial direction of the user is conceivably not agreeing with the direction in which recording is desired. That is to say, during the period from the facial direction of the user starting to change, and then the face of the user being directed toward the display device **800**, and the facial direction of the user changing again, and stopping at the original direction (hereinafter referred to as “action period”), the facial direction of the user does not agree with the direction in which the user desires to perform recording.

(421) Accordingly, the image clipping and developing processing unit **50** of the camera main unit **1** performs clipping and recording (developing) in the facial direction that was held at the time of the user starting to change his/her facial direction, during the action period for viewing the display device **800**. That is to say, the image clipping and developing processing unit **50** changes the clipping range set with respect to the frame image during the action period, during which the user is performing the action of directing his/her face toward the display device **800**, to the clipping range in the frame image before starting the action period.

(422) Whether the user has directed his/her face toward the display device **800** is determined by the display device control unit **801** of the display device **800**. In a case of determining that the user has directed his/her face toward the display device **800**, the display device control unit **801** notifies the camera main unit **1** that the user has directed his/her face toward the display unit **803** of the display device **800**.

(423) The display device control unit **801** can determine that the user has directed his/her face toward the display device **800** in a case in which the display unit **803** switches from an inactive state such as an unlit state, to an active state such as a lit state, and is activated, for example.

(424) Also, the display device control unit **801** is not limited to determination in accordance with switching from an unlit state to a lit state, and may determine that the user has directed his/her face toward the display device **800** by the user being image-captured (detected) by the front camera **805** that the display device **800** is provided with.

(425) Also, the display device control unit **801** may determine that the user has directed his/her face toward the display device **800** by a screen of an application for confirmation of the state of the camera main unit **1**, such as the screen described with reference to FIG. **13**, being displayed. The screen of the application may be displayed on the display unit **803** by operations performed by the user, or may be displayed on the display unit **803** by an instruction from the camera main unit **1**.

(426) Description will be made below regarding processing in FIG. **7A** according to the second embodiment that differs from the first embodiment. FIG. **20** is a flowchart of preparatory operation processing according to the second embodiment. The preparatory operation processing shown in FIG. **20** exemplifies detailed processing of step **S100** in FIG. **7A**. The preparatory operation processing according to the second embodiment is processing in which steps **S110** and **S111** are added to the preparatory operation processing in FIG. **7B** according to the first embodiment.

(427) Upon the power of the camera main unit **1** being turned on and the settings being switched on the basis of the image-capturing mode (steps **S101** to **S109**), the general control CPU **101** establishes a connection with the display device **800** by small power wireless in step **S110**. In step **S111**, the general control CPU **101** sends an instruction by small power wireless to turn off the lighting of the screen of the display device **800**. The display device control unit **801** of the display device **800** receives the instruction for turning the lighting of the screen off, and changes the state of the screen (display unit **803**) to the unlit state.

(428) The flow of processing at the display device **800** will be described with reference to FIG. **21**. The processing of each step in FIG. **21** is realized by the display device control unit **801** of the display device **800** controlling the components of the display device **800** in accordance with programs stored in the built-in nonvolatile memory **812**.

(429) In step **S521**, the display device control unit **801** stands by until the power switch is turned on and the power of the display device **800** goes on. In step **S522**, the display device control unit **801** establishes a connection with the camera main unit **1**. The display device control unit **801** can connect to the camera main unit **1** by small power wireless, for example. The processing in step **S522** corresponds to the processing of the camera main unit **1** in step **S110** described with reference to FIG. **20**.

(430) In step **S523**, the display device control unit **801** stands by until there is an instruction from the camera main unit **1** to turn the lighting of the screen off in step **S111** in FIG. **20**. Upon receiving the instruction to turn the lighting of the screen off, the display device control unit **801** advances to step **S524**. In step **S524**, the display device control unit **801** turns the lighting of the screen of the display device **800** to the unlit state.

(431) In step **S525**, the display device control unit **801** determines whether or not the display device **800** has been activated by operations performed by the user. In a case of determining that the display device **800** has been activated, the display device control unit **801** advances to step **S526**, and in a case of determining that activation has not been performed, advances to step **S527**. In step **S526**, the display device control unit **801** notifies the camera main unit **1** that the display device **800** has been activated.

(432) Note that although determination is made here regarding whether or not the display device **800** has been activated, it is sufficient for the display device control unit **801** to be able to determine whether or not the face of the user is directed toward the display device **800**. The display device control unit **801** may determine that the user has directed his/her face toward the display device **800** by the user being image-captured (detected) by the front camera **805** that the display device **800** is provided with. The method of determining whether or not the user has been image-captured may be any method as long as a face of a person or the face of the user him/herself can be detected in the image-captured image from the front camera **805**. For example, the display device control unit **801** can determine that the user has been image-captured in a case in which the face, or part of the face, of the user registered for facial authentication, is detected from the image-captured image of the front camera **805**.

(433) Also, the display device control unit **801** may determine that the user has directed his/her face toward the display device **800** by a screen of an application for confirmation of the state of the camera main unit **1**, such as the screen described with reference to FIG. **13**, being displayed.

(434) In step **S527**, the display device control unit **801** determines whether or not there has been notification from the camera main unit **1** that image-capturing has ended. In a case of not receiving the notification of image-capturing ending, the display device control unit **801** returns to step **S525**, and repeats the processing of steps **S525** and **S526** until the notification of image-capturing ending is received. In a case of receiving the notification of image-capturing ending, the display device control unit **801** advances to step **S528**.

(435) In step **S528**, the display device control unit **801** establishes a connection with the camera main unit **1** by high-speed wireless. In step **S529**, the display device control unit **801** receives a picture file, in which developing processing of a clipping range has been executed, from the camera main unit **1**. The display device control unit **801** executes the processing of steps **S800** to **S1000**, described with reference to FIG. **7A** in the first embodiment, with respect to the picture file that is received.

(436) Following the preparatory operation processing in FIG. **20**, the general control CPU **101** of the camera main unit **1** executes the facial direction detection processing (step **S200**), and the recording direction and range deciding processing (step **S300**), described with reference to FIG. **7A** in the first embodiment, and advances to the shooting processing of step **S400**.

(437) Processing for determining the action period of the user performing the action for viewing the display device **800** in the shooting processing in step **S400** will be described with reference to FIG. **22**. The processing shown in FIG. **22** is executed for each frame of the moving image that is image-captured by the shooting unit **40** in step **S400**. A frame that is the object of the processing in FIG. **22** will be referred to as “current frame”. The processing of each step in FIG. **22** is realized by the general control CPU **101** controlling the components of the camera main unit **1** in accordance with programs stored in the built-in nonvolatile memory **102**.

(438) In step **S401**, the general control CPU **101** determines whether or not a change-ongoing flag is on in the image-capturing of the frame preceding the current frame (hereinafter referred to as “preceding frame”). The change-ongoing flag is a flag indicating whether or not the user is performing a series of actions for viewing the display device **800**. The initial value of the change-ongoing flag is set to off, indicating that the user is not performing actions for viewing the display device **800**. The value of the change-ongoing flag is recorded in the primary memory **103** in step **S420**, for each frame.

(439) In a case of determining in step **S401** that the change-ongoing flag is off in the preceding frame, the general control CPU **101** advances to step **S402**. In step **S402**, the general control CPU **101** determines whether or not the face of the user is performing actions. Whether or not the face of the user is performing actions can be detected by change in the facial direction. Change in the facial direction can be detected as change in a directional vector of the facial direction detected in FIG. **7C** (observation direction v_i), or change in the position and the size of the picture recording frame (clipping range) decided on the basis of the directional vector, for example. In a case of detecting change of the directional vector, the general control CPU **101** can determine that the face of the user is performing actions in a case in which the amount of change of the directional vector is no smaller than a threshold value angle.

(440) In a case of detecting change in the clipping range, the general control CPU **101** acquires the position and the size (clipping range) of the picture recording frame recording in the primary memory **103** in the recording direction and range deciding processing (step **S300**) for the current frame and the preceding frame.

(441) The general control CPU **101** determines whether or not the difference in clipping ranges between the current frame and the preceding frame (e.g., distance between the position of the clipping range of the current frame and the position of the clipping range of the preceding frame) is

no smaller than a threshold value. The general control CPU **101** can determine that the face of the user is performing actions in a case in which the difference in clipping ranges between the current frame and the preceding frame is no smaller than the threshold value. On the basis of blurring (amount of change) of the facial direction in a case in which the user is not intentionally moving his/her face, the threshold value can be set as the greatest value, the average value, or the like, of this amount of change, for example. In a case of determining that the face of the user is performing actions, the processing advances to step **S403**. In a case of determining that the face of the user is not performing actions, the processing advances to step **S420**.

(442) In step **S403**, the general control CPU **101** sets the change-ongoing flag and an initial-action-ongoing flag to on, and sets a changing frame count to 1. An initial action is an action of the user directing his/her facial direction from the subject toward the display device **800**, before the user views the display device **800**. The initial-action-ongoing flag is a flag indicating a state in which the facial direction of the user is changing toward the display device **800** before the user views the display device **800**, and the setting thereof is carried on from the preceding frame.

(443) Even when the facial direction changes, this is not necessarily an action to view the display device **800**, so the change-ongoing flag is tentatively set to on. In a case in which determination is made that the change in facial direction is not an action to view the display device **800**, the change-ongoing flag is changed to off in step **S411**.

(444) The changing frame count is a variable for counting the number of frames from the facial direction of the user starting to change up to receiving the notification of activation from the display device **800**. In a case of not receiving notification of activation from the display device **800** for a predetermined period, it is conceivable that the user was not trying to confirm the state of the camera main unit **1** using the display device **800**, but changed the direction of his/her face to change the image-capturing direction. In a case of receiving notification of activation from the display device **800** while the changing frame count is smaller than the threshold value, the general control CPU **101** determines that the change in the facial direction was an action to view the display device **800**. Also, in a case in which the changing frame count reaches or exceeds the threshold value without reception of notification of activation from the display device **800**, the general control CPU **101** determines that the change in the facial direction was not an action to view the display device **800**. In step **S403**, the current frame is the first frame in which the face of the user starts to move, and accordingly the changing frame count is set to 1.

(445) In a case of determining in step **S401** that the change-ongoing flag of the preceding frame is on, the general control CPU **101** advances to step **S404**. In step **S404**, the general control CPU **101** determines whether or not the changing frame count is 0. In a case of determining that the changing frame count is not 0, the processing advances to step **S405**.

(446) In step **S405**, the general control CPU **101** determines whether or not the changing frame count is no smaller than the threshold value, and if smaller than the threshold value, advances to step **S406**. In a case in which the changing frame count is smaller than the threshold value, determination is made that the change in facial direction is an action for the user to view the display device **800**. In a case in which the changing frame count is not smaller than the threshold value, determination is made that the change in facial direction is not an action for the user to view the display device **800**.

(447) The threshold value for the changing frame count can be decided by, for example, multiplying a threshold value time from the user starting the action period for viewing the display device **800** up to reception of the notification of activation, by the framerate. The threshold value time may be around three to five seconds or so, for example, and may be optionally set by the user.

(448) In a case in which the notification of activation is received from the display device **800** within the threshold value time by the determination in step **S405**, the general control CPU **101** can change the clipping range of frames during the action period to the clipping range before moving the face. Also, in a case in which the notification of activation is not received from the display

device **800** within the threshold value time, the general control CPU **101** determines that the change in facial direction was not an action to view the display device **800**, and does not change the clipping range.

(449) In step **S406**, the general control CPU **101** determines whether or not the face of the user is performing actions, in the same way as in step **S402**. In a case of determining that the face of the user is not performing actions, the general control CPU **101** changes the initial-action-ongoing flag to off in step **S407**, and in a case of determining that the face of the user is performing actions, leaves the initial-action-ongoing flag on without being changed, and advances to step **S408**.

(450) In step **S408**, the general control CPU **101** determines whether or not the notification of activation has been received from the display device **800** between the image-capturing of the preceding frame and the image-capturing of the current frame. In a case of receiving the notification of activation from the display device **800**, in step **S410** the general control CPU **101** sets the changing frame count to 0. In a case of not receiving the notification of activation from the display device **800**, in step **S409** the general control CPU **101** increments the changing frame count. Upon setting the changing frame count in either step **S409** or step **S410**, the general control CPU **101** advances to step **S420**.

(451) In a case in which the changing frame count is no smaller than the threshold value in step **S405**, the general control CPU **101** determines that the change in facial direction is not an action for the user to view the display device **800**, and the processing advances to step **S411**. In step **S411**, the general control CPU **101** sets the change-ongoing flag and the initial-action-ongoing flag to off, and returns the changing frame count to 0.

(452) In step **S412**, the general control CPU **101** rewrites the change-ongoing flag, which had been continuously set to on up to the preceding frame, to off. That is to say, in a case in which the display device **800** does not go to an active state by the time that the number of frames image-captured from when the facial direction of the user starts to change reaches the threshold value, the general control CPU **101** determines that the actions of the user were not a series of actions for viewing the display device **800**. Accordingly, the general control CPU **101** rewrites the change-ongoing flag for each frame, which had been set to on from the time of the facial direction of the user starting to change, to off.

(453) In a case in which the changing frame count is determined to be 0 in step **S404**, the general control CPU **101** advances to step **S413**, and determines whether or not the face of the user is performing actions, in the same way as in steps **S402** and **S406**. In a case of determining that the face of the user is not performing actions, the general control CPU **101** advances to step **S414**.

(454) In step **S414**, the general control CPU **101** determines whether or not the initial-action-ongoing flag is on. In a case in which the initial-action-ongoing flag is on, the general control CPU **101** advances to step **S417**, and changes the initial-action-ongoing flag to off. That is to say, in a case in which the facial direction starts to change and is directed toward the display device **800**, and the change in facial direction stops, i.e., the initial action ends, before the user views the display device **800**, the general control CPU **101** sets the initial-action-ongoing flag to off.

(455) In a case of determining in step **S413** that the face of the user is performing actions, the general control CPU **101** advances to step **S418**, and determines whether or not the initial-action-ongoing flag is on. In a case in which the initial-action-ongoing flag is on, the general control CPU **101** advances to step **S420**, and in a case in which the initial-action-ongoing flag is off, advances to step **S419**.

(456) In step **S419**, the general control CPU **101** sets an ending-action-ongoing flag to on and advances to step **S420**. An ending action is an action in which, after viewing the display device **800**, the user directs his/her facial direction away from the display device **800** and toward the subject. The ending-action-ongoing flag is a flag indicating the state from the facial direction of the user starting to move again after viewing the display device **800**, up to stopping thereof, and the setting thereof is carried on from the preceding frame.

(457) In a case in which the initial-action-ongoing flag is not on in step **S414**, the general control CPU **101** advances to step **S415**, and determines whether or not the ending-action-ongoing flag is on. In a case in which the ending-action-ongoing flag is off, the general control CPU **101** advances to step **S420**, and in a case in which the ending-action-ongoing flag is on, the general control CPU **101** advances to step **S416**. In step **S416**, the general control CPU **101** determines that the ending action has ended, sets the ending-action-ongoing flag, and the change-ongoing flag of the current frame, to off, and advances to step **S420**.

(458) In step **S420**, the general control CPU **101** records, for each frame, the information of the clipping range, and the setting of the change-ongoing flag, in the primary memory **103**, as frame management information. FIG. **23** is a diagram conceptually exemplifying frame management information after moving image image-capturing up to frame No. $j+2$.

(459) The frame management information includes information of the frame No., the clipping range, and the change-ongoing flag. The frame No. is a number indicating what number in the moving image the frame is. The clipping range is indicated by the position and size of the picture recording frame with respect to this frame, for example. The change-ongoing flag is a flag indicating whether or not the user is performing an action for viewing the display device **800** (changing the facial direction) when performing image-capturing of this frame.

(460) Note that the frame management information is stored in the primary memory **103** separately from the picture file (moving image), but may be stored as metadata of the picture file. In this case, the frame meta described with reference to FIG. **15** further has an item of change-ongoing flag, and the values of the clipping range (clipping position) and the change-ongoing flag are saved in step **S420** as metadata of the picture file, in the same way as with the frame management information.

(461) The states of the changing frame count, the change-ongoing flag, the initial-action-ongoing flag, and the ending-action-ongoing flag, during the action period in which the user was performing actions to direct his/her face toward the display device **800**, will be described with reference to FIG. **24**. The states of the various types of flags will be described, assuming a flow in which the user directs his/her face toward the display device **800** to confirm the state of the camera main unit **1** and displays the confirmation screen, and after ending confirmation, returns his/her facial direction to the original position. FIG. **24** shows the states of the actions of the user, recording of facial direction, the various types of flags used in the processing in FIG. **22**, and the changing frame count, in time sequence.

(462) The user starts shooting (**A1**), and in a case of confirming the state of the camera main unit **1** for example, starts changing actions of the facial direction to view the display device **800** (**A2**). The change-ongoing flag is off until the changing actions of the facial direction start, and accordingly the processing of FIG. **22** advances from step **S401** to step **S402**. At the frame in which the changing actions of the facial direction start, in step **S403** the general control CPU **101** sets the change-ongoing flag and the initial-action-ongoing flag to on, and sets the changing frame count to 1.

(463) In each frame from the start of the changing actions of the facial direction until the display device **800** is activated (**A4**), the changing frame count is incremented by the processing of steps **S404** to **S409**. During this time, the change-ongoing flag is in the on state.

(464) When the changing actions of the facial direction for viewing the display device **800** stop (**A3**), the initial-action-ongoing flag is set to off in step **S407**. Upon receiving a notification of activation from the display device **800** (**A4**), the changing frame count is initialized to 0 in step **S410**.

(465) Note that in a case in which the changing frame count is no smaller than the threshold value in step **S405**, the changing actions of the facial direction are determined to not be actions for viewing the display device **800**, even if the notification of activation of the display device **800** is received. In this case, the change-ongoing flag recorded as on in the frames after start of the changing actions of the facial direction (**A2**) is rewritten to off in the frame management

information in FIG. 23, in step S412.

(466) In a case in which changing actions of the facial direction are started again (A5) after the user ends confirmation of the screen of the display device **800**, the processing in FIG. 22 advances in the order of steps S404, S413, and S418. The initial-action-ongoing flag is set to off when starting re-changing actions of the facial direction (A5), and accordingly the ending-action-ongoing flag is set to on in step S419.

(467) When the re-changing actions of the facial direction stop (A6), the processing in FIG. 22 advances in the order of steps S404, S413, S414, and S415. The ending-action-ongoing flag is set to on when the re-changing actions of the facial direction stop (A6), and accordingly the ending-action-ongoing flag and the change-ongoing flag are set to off in step S416.

(468) Recording range developing processing according to the second embodiment will be described with reference to FIG. 25. The recording range developing processing according to the second embodiment is detailed processing of step S500 in FIG. 7A. In the second embodiment, the recording range developing processing shown in FIG. 25 is executed instead of the recording range developing processing in FIG. 7E described in the first embodiment. The recording range developing processing according to the second embodiment differs from the recording range developing processing in FIG. 7E with respect to the point that the processing of step S502, in which the part of the picture recording frame **127i** is clipped from the ultrawide angle picture, is not executed.

(469) In the recording range developing processing according to the second embodiment, developing processing is executed on the entire region of the ultrawide angle picture read in step S501, and accordingly, the computation amount and the usage amount of the primary memory **103** increase. In a case in which the primary memory **103** is insufficient, the general control CPU **101** may perform recording to the built-in nonvolatile memory **102**.

(470) Also, an arrangement may be made in which the general control CPU **101** retains the processing of step S502, and clips a range that is larger than the picture recording frame **127i** described in the first embodiment, thereby limiting the computation amount and the usage amount of the primary memory **103** as compared to a case of developing the entire region. In this case, the general control CPU **101** can change the clipping range in the action period for viewing the display device **800**, within the range of the picture clipped as part of the entire range.

(471) In a case of clipping a range that is larger than that in the first embodiment in step S502, a facial direction that is assumed for the when the user views the display device **800** may be set in advance, and the clipping range may be decided so as to include the actual observation direction v_i and the picture in the facial direction that is set.

(472) The processing of steps S200 to S500 in FIG. 7A according to the second embodiment is repeatedly executed until recording ends. Recording range finalization developing processing that is executed after the recording range developing processing of step S500 ends will be described with reference to FIG. 26. The processing of each step in FIG. 26 is realized by the general control CPU **101** controlling the components of the camera main unit **1** in accordance with programs stored in the built-in nonvolatile memory **102**.

(473) In step S510, the general control CPU **101** notifies the display device **800** that image-capturing has ended. In step S511, the general control CPU **101** reads out information of one frame worth of the frame management information described with reference to FIG. 23 from the primary memory **103**. The general control CPU **101** sequentially reads out the frame management information from the first frame, and repeats the processing of steps S512 to S516 for the frames that are read out.

(474) In step S512, the general control CPU **101** determines whether or not the change-ongoing flag is on. In a case in which the change-ongoing flag is not on, the general control CPU **101** advances to step S513, and executes the clipping processing for the clipping range that is read out. In step S514, the general control CPU **101** holds the clipping range (X_i, Y_i, WX_i, WY_i) clipped in

step S513. The clipping range that is held is used for changing the clipping range during the action period in which the user views the display device **800**, as the clipping range immediately before the change-ongoing flag changing to on.

(475) In a case in which the change-ongoing flag is on in step S512, the general control CPU **101** advances to step S515, and changes the clipping range of the frame that is currently the object of processing to the clipping range (Xi, Yi, WXi, WYi) held in step S514. In step S516, the general control CPU **101** executes clipping processing in the clipping range set in step S515.

(476) In step S517, the general control CPU **101** determines whether or not clipping processing has ended for all frames, and in a case in which this has not ended, repeats the processing of steps S511 to S516 until the last frame.

(477) A case of applying the processing in FIG. **26** to the frame management information in FIG. **23** will be described. The change-ongoing flag is off at frame **1** to frame **i**, and accordingly the general control CPU **101** advances from step S512 to step S513. Upon the processing of step S514 being executed with respect to frame **i**, the clipping range that is held is (Xi, Yi, WXi, WYi).

(478) At frame **i+1**, the change-ongoing flag is on, and accordingly the general control CPU **101** advances to step S515. The general control CPU **101** sets the clipping range of frame **i+1** to (Xi, Yi, WXi, WYi), and executes clipping processing. The change-ongoing flag is on at frame **i+2** to frame **j+2** as well, and accordingly the general control CPU **101** sets the clipping range of frame **i+2** to frame **j+2** to (Xi, Yi, WXi, WYi) in the same way as with frame **i+1**, and executes clipping processing.

(479) When the recording range finalization developing processing in FIG. **26** ends, the general control CPU **101** (generating means) executes the processing of steps S600 to S1000 in FIG. **7A** in the same way as in the first embodiment.

(480) According to the second embodiment above, the camera main unit **1** can perform clipping in the facial direction in which the face of the user was directed when the facial direction started to change, from the user starting image-capturing through the course of the action period for viewing the display device **800**. Accordingly, even when the user directs his/her line of sight away from the direction of the subject, and directs his/her facial direction toward the display device **800** and confirms the state of the camera main unit **1** during image-capturing of the moving image, the user can continue to record a desired picture without interrupting image-capturing.

(481) Although preferred embodiments of the present invention have been described above, the present invention is not limited to these embodiments, and various modifications and alterations may be made without departing from the scope and spirit thereof.

(482) According to the present disclosure, image-capturing of desired pictures can be performed even when the user directs his/her line of sight away from the direction of a subject while performing image-capturing of a moving image.

Other Embodiments

(483) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable

instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(484) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(485) This application claims the benefit of Japanese Patent Application No. 2022-063909, filed on Apr. 7, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image-capturing apparatus, comprising: an image-capturing device; and at least one memory and at least one processor which function as: a detecting unit configured to detect a facial direction of a user with respect to the image-capturing apparatus; a setting unit configured to set a clipping range in each frame image in a moving image that is image-captured by the image-capturing device, on the basis of the facial direction that is detected; and a generating unit that is configured to generate a clipped moving image from the clipping range, wherein the generating unit generates the clipped moving image by changing the clipping range set with respect to a frame image in an action period, during which the user performs an action of directing a face toward a display device communicably connected to the image-capturing apparatus, to the clipping range set with respect to a frame image from before the action period starting.
2. The image-capturing apparatus according to claim 1, wherein the action period is a period from the facial direction of the user starting to change, up to the face of the user being directed toward the display device and then the facial direction of the user changing again and stopping.
3. The image-capturing apparatus according to claim 1, wherein the generating unit generates the clipped moving image by changing the clipping range set with respect to the frame image in the action period to the clipping range set with respect to the frame image from before the action period starting, in a case of receiving, within a threshold value time from the action period starting, a notification from the display device that the user has directed the face toward the display device.
4. The image-capturing apparatus according to claim 1, wherein the generating unit generates the clipped moving image by changing the clipping range set with respect to the frame image in the action period to the clipping range set with respect to the frame image from before the action period starting, in a case a count of frames image-captured from the action period starting, up to receiving a notification from the display device that the user has directed the face toward the display device, is smaller than a threshold value.
5. A control method of an image-capturing apparatus, the control method comprising: performing image-capturing; detecting a facial direction of a user with respect to the image-capturing apparatus; setting a clipping range in each frame image in a moving image that is image-captured in the image-capturing, on the basis of the facial direction that is detected; and generating a clipped moving image from the clipping range, wherein, in the generating, the clipped moving image is generated by changing the clipping range set with respect to a frame image in an action period, during which a user performs an action of directing a face toward a display device communicably connected to the image-capturing apparatus, to the clipping range set with respect to a frame image from before the action period starting.
6. A non-transitory computer-readable medium that stores a program for causing a computer to execute a control method of an image-capturing apparatus, the control method comprising: performing image-capturing; detecting a facial direction of a user with respect to the image-

capturing apparatus; setting a clipping range in each frame image in a moving image that is image-captured in the image-capturing, on the basis of the facial direction that is detected; and generating a clipped moving image from the clipping range, wherein, in the generating, the clipped moving image is generated by changing the clipping range set with respect to a frame image in an action period, during which a user performs an action of directing a face toward a display device communicably connected to the image-capturing apparatus, to the clipping range set with respect to a frame image from before the action period starting.

7. A display device that is communicable with the image-capturing apparatus according to claim 1, the display device comprising at least one memory and at least one processor which function as: a determining unit configured to determine whether or not the user has directed the face toward the display device while performing image-capturing by the image-capturing device; and a notifying unit configured to, in a case of determining that the user has directed the face toward the display device, notify the image-capturing apparatus that the user has directed the face toward the display device.

8. The display device according to claim 7, wherein the determining unit determines that the user has directed the face toward the display device in accordance with one of the display device switching from an unlit state to a lit state, image-capturing of the user being performed by a front camera included in the display device, and a screen of an application for confirming a state of the image-capturing apparatus being displayed.

9. An image-capturing system having an image-capturing apparatus and a display device that is communicable with the image-capturing apparatus, the image-capturing apparatus comprising: an image-capturing device; and at least one memory and at least one processor which function as: a detecting unit configured to detect a facial direction of a user with respect to the image-capturing apparatus; a setting unit configured to set a clipping range in each frame image in a moving image that is image-captured by the image-capturing device, on the basis of the facial direction that is detected; and a generating unit that is configured to generate a clipped moving image from the clipping range, wherein the generating unit generates the clipped moving image by changing the clipping range set with respect to a frame image in an action period, during which the user performs an action of directing a face toward the display device communicably connected to the image-capturing apparatus, to the clipping range set with respect to a frame image from before the action period starting, and the display device comprising at least one memory and at least one processor which function as: a determining unit configured to determine whether or not the user has directed the face toward the display device while performing image-capturing by the image-capturing device; and a notifying unit configured to, in a case of determining that the user has directed the face toward the display device, notify the image-capturing apparatus that the user has directed the face toward the display device.
